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EV Charging Load Demand Prediction Using Machine Learning in a SUMO-Based Simulation Environment

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This is to certify that **Rakesh Biswas** (ID: 200112) has successfully completed the B.Sc. thesis titled *"EV Charging Load Demand Prediction Using Machine Learning in a SUMO-Based Simulation Environment"* at the Department of Computer Science and Engineering, Jashore University of Science and Technology, Jashore-7408, Bangladesh, in May 2026.

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Dedication

To my beloved parents, whose endless love, sacrifice, and encouragement have been the cornerstone of every achievement in my life.

To my teachers, who ignited the flame of curiosity within me and guided me towards the pursuit of knowledge.

And to all the researchers and engineers working tirelessly towards a sustainable, electrified future of transportation.

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Abstract

The rapid growth of Electric Vehicle (EV)s presents both an important opportunity and considerable challenge for modern power grid management. To make optimize energy distribution, accurate prediction of EV charging load demand is really important. It decreases unbalanced load stress on electrical infrastructure and enables smart grid operations.

This thesis presents a comprehensive framework for predicting EV charging load demand using a suite of supervised Machine Learning (ML) and Deep Learning (DL) models applied to data generated from a realistic microscopic traffic simulation. To enable real-time vehicle monitoring and intelligent charging station management, the simulation environment was built by integrating the open-source urban mobility simulator SUMO with the Traffic Control Interface (TraCI). A network of 10 bidirectional charging stations was deployed with configurable power ratings ranging from 25 kW to 50 kW. Also, separate charging and waiting queue slots were configured for each station. Unique electric vehicles of the battery-powered type were simulated for 600 seconds. This resulted in 34,020 vehicle-level records, including features such as state of charge (SOC), speed, acceleration, station utilization, queue length and system-wide load metrics. Several different models were evaluated for system-level load demand prediction.

In addition, a rule-based smart charging recommendation system was implemented. In this scenario, vehicles were routed to the optimal charging station based on real-time SOC level, distance to the nearest station, slot availability and queue status. The findings show that SUMO-based simulation is an effective method for generating enriched EV charging datasets.

Keywords: Electric Vehicle, Charging Load Prediction, Machine Learning, SUMO Simulation, Artificial Neural Network (ANN), Random Forest, XGBoost, Smart Charging, SOC, Queue Management.

Chapter 1

Introduction

Chapter 1: Introduction

The global transportation sector is undergoing a radical change due to the urgent need to reduce greenhouse gas emissions and to reduce dependence on fossil fuels. For totally decarbonizing road transport, Electric Vehicle (EV)s have emerged as one of the most promising technologies. According to the [1], the number of electric vehicles worldwide exceeded 26 million in 2022 and is forecast to continue to grow rapidly in the coming decades. With the rapid growth of electric vehicle usage, the increasing demand for electricity for charging is becoming a serious concern for both power system managers and urban planners [2]. Recently, advanced integrated frameworks and deep learning hybrids have been proposed to address the increasingly complex nature of these load patterns [3, 4].

The number of vehicles or EVs is growing really fast. This means we need systems to control energy use so that we can use energy in the best way possible. In countries like Bangladesh we see a lot of three-wheeled vehicles like Easy Bikes, Electric Rickshaws and Electric Vans. These electric vehicles are very important for people to move around in cities and towns.. When people charge these electric vehicles it can be a problem because they do not charge them at the same time [5]. This can cause changes in how much electricity is being used, which can be very hard on the power grid. Electric vehicles or EVs are used a lot in Bangladesh and other places so we need to find ways to charge them that will not cause problems, for the power grid.

Electric vehicles EVs communicate directly with the power grid. When lots of vehicles charge at the same time especially during peak hours it can cause big problems for the distribution network. This can lead to voltage violations and transformer overloads. Also it can increase energy costs [6]. Electric vehicles can do this because they use a lot of power. Therefore we need to manage vehicle charging loads in a good way. We need tools that can predict when and where electric vehicles will charge. This will help us understand the demand patterns, across the vehicle charging station network. We need to know this so we can plan for vehicles[7].

In this field, one of the key challenges is to accurately predict system-level charging load demand. Reliable load prediction enables better planning of charging stations, improved power distribution and the prevention of overload situations.

1.1 Background

The rapid increasing of electric vehicles is reshaping transportation and electricity distribution systems. Unlike conventional fueled vehicles, electric vehicles, shift energy demand from fuel stations to the power grid . This creates loads that vary over time and space. Understanding the movement, energy consumption and charging behavior of electric vehicles is essential for load prediction, grid reliability and optimal utilization of charging infrastructure [7]. In cities of Bangladesh , a major drawback is the unorganized and untested charging behavior of electric vehicles . Vehicles oftenly crowd around specific nodes or roadside charging points. Which is resulting in large amounts of energy consumption during peak hours . This often results in stress on the local grid, voltage drops and overloading of transformers problems. Which that already exist in rapidly growing cities like Khulna, Jashore, Satkhira. Besides, energy consumption, charging-load varies greatly depending on vehicle speed, congestion levels, route selection, and driver behavior . Without proper modeling and prediction of these parameters, city authorities and electricity distributors face increasing difficulties in ensuring grid reliability, planning future charging stations and controlling unbalanced load distribution. This research work uses traffic simulation tools such as SUMO to model real-world urban mobility. Where EV extensions to SUMO help model battery behavior, charging events, energy consumption, system load behavior. These outputs are integrated with load prediction models to create a robust approach for analyzing the load impact of EV adoption.

1.1.1 Definition

1.1.1.1 Battery-Powered Vehicle (BPV) or EV

A Battery Powered Vehicle(BPV) or Electric Vehicle (EV) is a transportation system which is powered entirely by rechargeable batteries and electric motors instead of an internal combustion engine. EVs store electrical energy in lithium-ion or lead-acid batteries and through an electric drivetrain, they convert this stored energy into mechanical energy [2]. In the context of Bangladesh in urban areas like Khulna and Jessore -electric vehicles include the widely used low-cost electric three-wheelers (e.g., easy bikes , e-rickshaws, e-vans). Those vehicles are popular due to their low operating costs, absence of exhaust emissions and suitability for short-distance travel within cities or urban area . Their electricity-based operations directly link transportation demand to local electrical grid-loads. Which is making EV behavior a critical element in grid planning.

(a) Easy Bike on the road

(b) Individual Easy Bike vehicle

(c) Electric Van

(d) Electric Rickshaw

Figure 1.1: Commonly used electric three-wheelers and battery-powered vehicles in Bangladesh.

1.1.1.2 Load Prediction

Load prediction refers to the process of estimating the total electrical demand of a system based on historical or observed data. In this study, it specifically refers to the prediction of the system-level charging load (`system_total_load_kW`) generated by electric vehicles using machine learning models [8]. The prediction is performed by understanding the relationship between various input characteristics —such as time progress , number of vehicles charging, system-level vehicle statistics and battery charge status and the corresponding load demand . Unlike traditional analytical approaches , this research uses data-driven models to understand the complex and nonlinear dependencies within the system. Proper load prediction can help to understand in better way how electric vehicle charging behavior affects overall energy demand and helps in efficient planning and management of charging infrastructure.

1.1.1.3 Traffic Microsimulation

Traffic microsimulation is a precise, vehicle-level modeling technique that which captures the detailed movements of individual vehicles over time. It represents the interaction of speed, acceleration, deceleration, lane change behavior, path selection, stop-and-go cycles and congestion with high temporal accuracy. Microsimulation becomes increasingly important in EV research. As vehicle speed directly affects the dynamics of energy consumption and battery degradation. By integrating microsimulation with EV models , researchers can generate realistic estimates of power consumption, load behavior under different road, time and traffic conditions. Which ultimately helps in more accurate charging load analysis and prediction .

1.1.1.4 State of Charge (SOC)

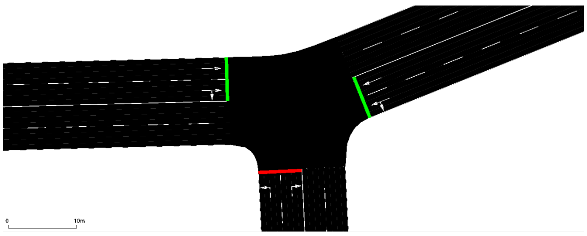
State of Charge (SOC) is a quantitative measurement which indicates the remaining energy stored in an electric vehicle (EV) battery [9]. It is usually expressed as a percentage (0–100%). SOC decreases when the vehicle is being driven due to the use of propulsion power and it increases when the vehicle is connected to a charging station. As it determines when and where a vehicle will need charging , SOC is a fundamental variable for understanding electric vehicle behavior . In simulation environments like SUMO, SOC-logs give : time-step-level insights into real-time power consumption. This enables the creation of detailed charging demand models and node-level load prediction profiles.

1.1.1.5 Charging Station

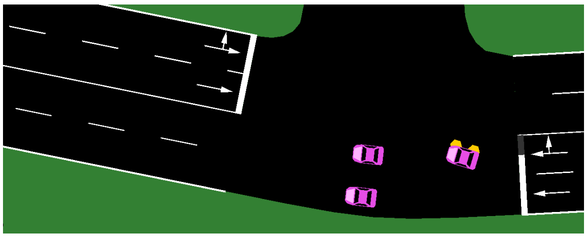
A charging station is a road-side power supply point where electric vehicles (EVs) recharge their batteries. In simulation-based studies, charging stations are modeled as infrastructure elements . Those are connected to specific network nodes or road segments. Each station has certain characteristics, such as : charging capacity, availability, deployment strategy and charging session duration.

1.1.1.6 Simulation of Urban Mobility (SUMO)

SUMO (Simulation of Urban Mobility) [10] is an open-source, microscopic traffic simulation platform. It is used for modeling and analyzing transportation networks. SUMO supports user-defined road networks, custom vehicle flows, routing algorithms, and mobility patterns. Through its EV extension, Sumo can simulate battery discharge, energy consumption, regenerative braking and real-time charging events. Sumo provides a controlled environment for research on the EV ecosystem in Bangladesh. Here in SUMO urban traffic behavior can be simulated and the resulting charging demand can be measured without the need for large-scale real-world experiments.



(a) NetEdit interface for road network design.



(b) SUMO simulation environment.

Figure 1.2: SUMO simulation tools: NetEdit and the main SUMO GUI.

1.1.1.7 Traffic Control Interface (TraCI)

The Traffic Control Interface (TraCI) [11] is SUMO's external programming interface. It allows real-time interaction between the simulation and external applications. TraCI enables dynamic control of vehicle characteristics, simulation variable extraction, traffic signal management and continuous logging of EV parameters such as SOC, speed, energy consumption and charging events. In EV load prediction research, TraCI plays an important role in automated data collection. For prediction and visualization it also integrates SUMO outputs with Python-based analytical models.

1.1.1.8 Power Consumption Analysis

Power consumption analysis refers to the study of how electric vehicles (EVs) use electricity across different driving conditions, speeds, accelerations and network paths. This includes quantifying energy consumption per route, per node and per unit distance [12]. this is the means by which the study will get to understand the stress points on the power grid network system. One thing that cannot be stressed enough is how important it is within the framework of urban settings such as those in Khulna and Jessore.

1.1.1.9 Load Forecasting Model

A load forecasting model is a computational method—often statistical or machine learning-based. It is used to predict future EV charging loads using past charging shortages, route flows , traffic congestion and charging station usage patterns. These models help predict upcoming power demand, optimize the placement of charging stations [7]. They also guide power suppliers in planning feeder capacity expansion. With the rapidly expanding use of electric vehicles in Bangladesh, such forecasts are essential to prevent transformer overloading and grid instability.

1.1.1.10 Urban EV Mobility Pattern

Urban electric vehicle mobility patterns refer to the characteristic movement behavior of electric vehicles within a city. It includes typical routes, peak congestion times, center-level activity , trip length and charging intervals. By understanding mobility patterns, researchers can link electric load distribution to traffic dynamics . In Bangladeshi cities —where electric vehicles like Easy Bikes, E-Rickshaws follow regular routes between markets, hospitals and government institutions. This analyzing mobility patterns becomes essential for accurate simulation modeling and load forecasting.

1.1.1.11 Queue Management and Promotion Logic

Queue management is an infrastructure-level process -which coordinates the flow of vehicles at a charging station. It distinguishes between active charging slots (where energy is transferred) and waiting slots (where vehicles are queuing for service). Promotion logic refers to the systematic process of moving a vehicle from a waiting state to a charging state as soon as a slot becomes available. This generally follows a first-in first-out (FIFO) protocol to make sure fairness and simulation accuracy [13].

1.1.1.12 Rule-Based Recommendation System

A rule-based recommendation system is a decision-making algorithm -which uses some predefined "if-then" conditional statements to suggest the best optimal actions. In the case of EV charging, this system evaluates real-time telemetry such as a vehicle's SOC, distance to the nearest station and current queue length to recommend the most suitable charging locations [14]. It also balances the overall load of the network with the needs of each vehicle.

1.1.1.13 Heterogeneous EV Fleet

A heterogeneous EV fleet refers to a collection of different types of electric vehicles within a single simulation or study. It is characterized by different physical sizes, battery capacities and power requirements. In urban modeling in Bangladesh, this primarily includes a mix of easy bikes, electric rickshaws and electric vans . It is essential to take this diversity into account to capture the complex and uneven energy usage patterns. This is typical of real-world traffic.

1.1.1.14 Spatiotemporal Demand Pattern

Spatiotemporal demand pattern refers to the distribution of electrical load, across both space (geographical location of charging nodes) and time (simulation period or daily interval) . By analyzing these patterns, researchers and service providers can identify charging 'hotspots' and peak demand times. Wchich, enabling more precise grid reinforcement and infrastructure planning.

1.2 Problem Statement

The fast increasing of electric vehicles (EVs), specially in urban transportation systems, has created significant challenges in accurately prediction the charging load demand on the electricity grid. In Bangladesh, where electric three-wheelers such as e-bikes and electric rickshaws are widely used [5], the electricity distribution infrastructure is often already under stress during peak hours. The addition of EV charging intensifies this challenge due to the highly dynamic and nonlinear charging behavior . Which is influenced by factors such as -vehicle movement, battery state of charge , charging station availability and vehicle distribution at the system-level. A major limitation of existing research is the lackings of comprehensive and large scale real world datasets . That capture both vehicle and station level features, especially in developing areas. Besides, many conventional and statistical methods fail to effectively model the complex relationships between these variables. Where some advanced models can be computationally expensive and less intuitive for practical application in resource-limited environments . So, there is a critical need for a data-driven, effective and understandable approach to system level EV charging load demand prediction. This study addresses those challenges by using a simulation-based dataset that includes detailed system and vehicle characteristics . It implementes and evaluates multiple machine learning models to accurately predict the overall load demand (*total system load kW*) under changing system conditions.

1.3 Objectives

The primary objectives of this research are follows :

1. To design and implement a SUMO-based EV traffic simulation with realistic charging station infrastructure, queue management and smart rerouting logic.

2. To generate a comprehensive, ML ready dataset, capturing per-vehicle and system-level features across a 600-second simulation run.

3. To train and evaluate seven supervised machine learning and deep learning models —Linear Regression (LR), Decision Tree (DT), Random Forest (RF), Extreme Gradient Boosting (XGB), Artificial Neural Network (ANN), Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU) —for predicting the system-level EV charging load demand.

4. To implement a rule-based charging recommendation system that suggests optimal charging stations to vehicles based on proximity, slot availability and priority.

5. To analyze model performance using standard regression metrics (Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Coefficient of Determination (R^2)) and identify the most suitable approach for this domain.

1.4 Research Questions

This study seeks to answer the following research questions :

1. Can a SUMO-based microscopic simulation environment generate a comprehensive and realistic dataset for modeling EV charging load demand?

2. Which machine learning model provides the most accurate prediction of system-level EV charging load (`system_total_load_kW`)?

3. What are the most influential features affecting EV charging load demand in a multi-vehicle system?

4. How effectively can machine learning models capture the relationship between system-level and vehicle-level features for accurate load prediction?

5. How does the performance of linear models compare with tree-based models for EV load prediction?

6. How accurately can the trained models predict load demand across different system conditions?

1.5 Significance of the Study

The significance of this research lies in contributing to the growing challenge of managing electric vehicle (EV) charging demand through a data driven and scalable approach. With the rapid growth of electric vehicles, specially in developing regions like Bangladesh [5] —power distribution systems are increasingly facing unpredictable and highly variable load patterns due to unbalanced charging behavior. This study provides a realistic framework for accurately predicting system-level EV charging load (`system_total_load_kW`) using a comprehensive simulation-based dataset . It incorporates both vehicle-level dynamics and system-level operational characteristics. By applying and comparing multiple machine learning models, this study shows that advanced yet in computationally inexpensive techniques, particularly tree-based models- can effectively capture the complex nonlinear relationships inherent in the EV charging ecosystem. The study results highlight that load demand is largely driven by a few key system-level factors. Which is enabling stakeholders to better understand the underlying determinants of energy consumption. It has a direct impact on grid management, which includes improved demand prediction, optimal power distribution, improved planning of charging infrastructure and peak load stress mitigation on transformers and substations. Moreover, the use of understandable models and feature importance analysis ensures that the results are not only accurate but also interpretable. This is making them suitable for application in resource limited environments. Overall , this research bridges the gap between simulation-based data generation and real-world applicability. And this is providing a strong foundation for future research and practical applications in intelligent energy management systems for EV-integrated smart grids.

Chapter 2

Literature Review

Chapter 2: Literature Review

The research field related to EV charging load prediction and smart charging management encompasses multiple disciplines, including power systems engineering, transportation science and computer science . This chapter reviews the most relevant works in three thematic areas: (i) EV charging load modeling and pediction, (ii) simulation-based methods for EV data generation and (iii) machine learning methods for energy load prediction.

2.1 EV Charging Load Modeling

Early approaches to EV charging load modeling dependent on probabilistic approaches, which used statistical distributions of arrival times, departure times and daily travel distances to generate artificial load profiles [15]. Sortomme et al. [6] showed that the coordinated charging strategies can significantly reduce distribution system losses compared to uncontrolled charging. He et al. [16] discussed the optimal placement of charging stations. Which showed that accessibility and driving range limitations play an important role in station usage patterns. More recent researches have adopted a data-driven approach. Chen et al. [12] integrated probabilistic load flow analysis with EV charging uncertainty to assess grid impact. This provides a basis for the feature engineering techniques. Which is also used in the present work. Ul-Haq et al. [15] modeled residential electric vehicle charging patterns and highlighted the importance of highlighting spatial variations between charging locations.

2.2 Machine Learning for Load Forecasting

ML has become the dominant paradigm for short-term electricity load prediction. Zhang et al. [7] proposed a complete deep learning framework for EV charging load forecasting. Which incorporated historical demand, time-of-day characteristics and weather data and achieved significant improvements over classical autoregressive models. Wang et al. [8] applied random forest (RF) regression in short term load forecasting at EV charging stations. That is demonstrating robustness against noisy sensor data and missing values. Gradient boosting methods have also achieved state-of-the-art results. Chen et al. [17] applied XGBoost to one day EV load forecasting That is exploiting its regularization ability to prevent overfitting on small datasets. Tayeb and Hyndman [18] compared gradient boosting with conventional time series methods. And they found that tree-based algorithms always performed better than ARIMA algorithms for nonlinear load behavior. While linear regression continues to be an important starting point for load forecasting. Guo et al. [19] showed that linear models can capture the dominant trends in EV charging demand when the features are carefully designed, specially when polynomial and interaction terms are included. More recently, Siddiqui et al. [3] demonstrated that by combining deep feature extraction with boosting techniques—integrated DeepBoost approach can further refine forecasting accuracy. Same way, Ran et al. [4] introduced a refined CNN-LSTM-AM model to specifically target multi-scale temporal dependencies in EV-charging data, setting new benchmarks for sequential load prediction.

2.3 SUMO-Based EV Simulation

SUMO has been widely adopted for EV research because of its open-source nature, extensibility and support for battery models. Lopez et al. [10] describe the core capabilities of SUMO, including models of electric vehicle energy consumption, parking areas and a Traffic Control Interface (TraCI) API for external control. Wegener et al. [11] introduced the TraCI interface, that enables real time monitoring and manipulation of simulated vehicles. This makes it ideal for implementing adaptive charging management algorithms . Helbing and Triber [20] established the theoretical foundations for microscopic car-following models. That are the basis of SUMO's vehicle dynamics. The integration of these dynamics with electric powertrain models allows for accurate simulation of energy consumption patterns. And this directly affects the state of charge (SOC) trajectory and charging events.

2.4 Smart Charging and Recommendation Systems

Yilmaz and Crain [2] gave a comprehensive review of vehicle-to-grid technologies, identifying intelligent charging coordination as a key enabler of grid stability . Ding et al. [13] proposed a two-stage robust optimization for distributed energy management. That concept is related to the queue-aware routing logic implemented in this work. For EV charging coordination, Nguyen et al. [14] developed an adaptive dynamic programming method . Which showed that real-time SOC aware routing can significantly reduce the average waiting time. This motivates the rule based recommendation engine developed in the present study. Which selects stations based on SOC priority level and slot availability.

2.5 Research Gaps

Despite significant work in this area, several gaps remain. Most existing research relies on real-world datasets that are difficult to obtain, especially in the context of developing countries [5]. The creation of simulation based datasets using SUMO with integrated queue management and real-time recommendation systems has not been widely explored. Moreover, comparative benchmarking of multiple ML models on simulated EV load data is limited in the literature . This thesis addresses these shortcomings through an integrated simulation, data generation and ML prediction framework.

Chapter 3

Methodology

Chapter 3: Methodology

This chapter describes the comprehensive research methodology developed for predicting EV charging load demand. The approach integrates microscopic traffic simulation with advanced machine learning techniques to capture the complex, stochastically driven dynamics of urban energy consumption [10]. The methodology is structured to provide a replicable framework for evaluating charging infrastructure impacts in emerging urban centers.

3.1 Research Design

This research follows a quantitative, simulation-based experimental design centered on the "Monitor–Predict–Control" paradigm. Which is designed to bridge the gap between microscopic dynamics patterns and grid level electrical load - demand. As shown in Figure 3.1, the research design is performed through a systematic multi stage workflow. The whole process starts with collecting data and establishing the environment in the simulation tool. With that set up, a **Road Network Designing** was carried out to simulate the features of the actual road network of the urban area of Bangladesh [5] (for example Khulna city). Then **Traffic Light System Placement** and **Charging Station Placement** were strategically placed to simulate realistic infrastructural limitation and constraints. After the establishment of the physical environment , the approach shifts to modeling by determining **Vehicle Configuration in SUMO** [10] and **Vehicle Fleet Characterization**, representing the operational behavior of electric three-wheeled vehicles.

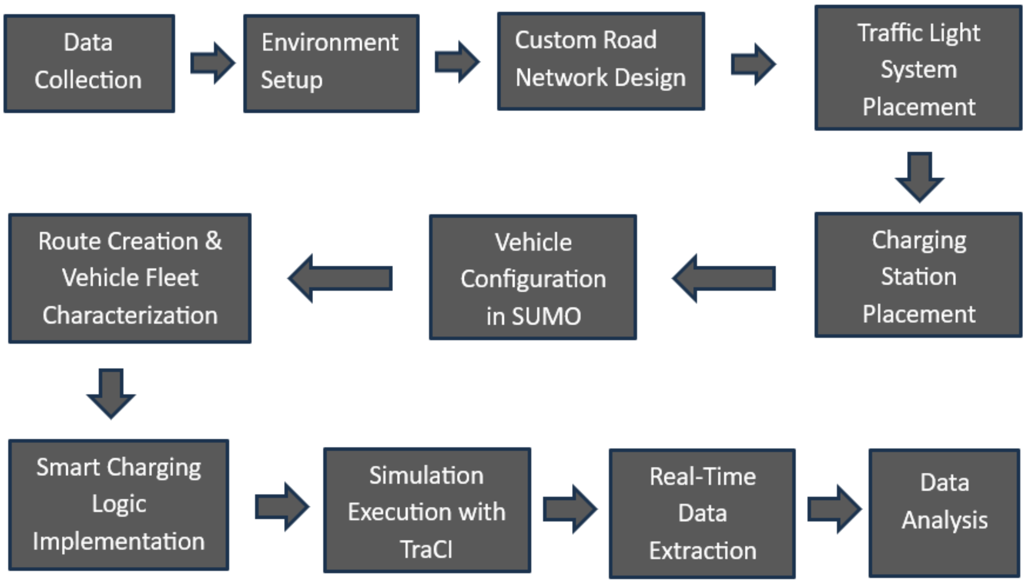


Figure 3.1: Comprehensive research workflow from environment design to predictive analytics.

In order to enhance the level of system's reality and make the system smarter, a queue management logic implementation [13] has been considered to manage the congestion of vehicles by taking into account various parameters such as the size of queues and their availability. Furthermore, **Smart Charging Logic Implementation** [2] has been added to the simulation system which makes vehicles go to different charging stations by considering various variables depending on the system states. The next step in our study is to execute the developed simulation with the help of **TraCI framework** [11]. In this phase, we have created a monitoring layer that helps extract data in real time during the execution of the simulation process. The extracted data include both the system-level and the vehicle-level features which describe their behavior and characteristics at every step of the process. Moreover, we can extend the scope of our research and consider the features connected to charging recommendations, although it is not the purpose of our research work currently. It can be useful for further in-depth study.

3.2 Data Collection

Data Collection Stage The data gathering stage was planned in such a way that the simulation system along with the machine learning algorithms were created taking into account practical operating attributes of electric three-wheeled vehicles extensively utilized in cities of Bangladesh [5]. At first, comprehensive information was gathered about three major types of battery-operated vehicles: Easy Bikes, Electric Rickshaws, and Electric Vans – which are reportedly the primary modes of electric transport in places such as Khulna. The objective of this stage was to incorporate both physical and behavioral aspects of the vehicles in order to simulate the dynamics of their movement. For this, specifications of the vehicles were gathered from multiple sources including manufacturer details, technical documentation, and input from local drivers/vehicle operators. This information was obtained on such parameters as vehicle dimensions (length, width, height), speed, acceleration and deceleration capabilities and all important characteristics of the power, for example, batteries capacity, its recharging ability, energy usage, and mobility. Also, we considered the practical use of these cars – how drivers interact with the vehicle, what kinds of routes they usually take and when they usually recharge. As a result, we have clearly understood the way in which vehicles interact with the charging stations in practice.

However, it did not only involve collecting information about the vehicles. Contextual parameters, such as the way people used the chargers, the time taken to charge and variations in demand during the different parts of the day, were also taken into consideration because those aspects helped build more realistic simulations and create a database, which would be both comprehensive and complete.

Once the data have been collected and validated, it is used to classify the type of vehicles as well as behavior rules in simulation models, which form the basis for accurate prediction of the demand for load. In the end, such extensive data collection makes it clear that the study is not theoretical; rather, it is practical, which allows it to survive in the real world and be applicable to others.

3.3 Implementation

This portion depicts the practical use of the research framework in technical terms. It includes setting up of the simulation environment, developing logic modules, and incorporation of data acquisition and machine learning modules. The primary objective of such an implementation is the actualization of the Monitor-Predict-Control system in order to simulate real EV behavior and acquire quality data for load predictions.

3.3.1 Environment Setup

In order to conduct the experiments for this research work, the environment has been set up in such a way that modeling of EV dynamics and their charging is done with utmost reliability under controlled but realistic settings. The experiments have all been performed using a 64-bit Windows 10-based machine.

The primary simulation environment employed in this research study is Simulation of Urban Mobility (SUMO), the Simulation of Urban Mobility v.1.25.0 [10]. This is an open-source microscopic traffic simulation package that is widely known for its capacity to simulate the behavior of individual automobiles and its inbuilt capability for electric vehicle mobility features, such as battery management [10].

The custom road network and simulation framework have been developed in the NetEdit application provided by SUMO, the graphical net-editor. This editor allows users to design and adjust anything from road configurations, junctions, traffic lights to locations of the EV charging stations. Thanks to the NetEdit editor, we managed to develop a realistic traffic environment model based on the layout in urban environments such as Khulna and Jashore city. To interact with the simulation in real-time, we employed the Traffic Control Interface (TraCI) (*Traffic Control Interface*) [11]. The TraCI interface facilitates direct control of the simulation process.

As our main programming language, Python 3.10.12 was chosen to manage the whole simulation, obtain data and do all the necessary computations. Utilizing TraCI, we implemented several scripts that were responsible for performing simulation, setting up queue management, allowing smart charging functions, and obtaining all required data at the runtime of simulation. Such data includes data about each vehicle individually and the entire traffic situation, which can be described by such parameters as: speed, State of Charge (SOC), number of charging vehicles, station occupancy, and total system load. Using `pandas` library, we managed to process such data, and then store it in well-structured file formats such as CSV or Excel.

In summary, such an integrated ecosystem provides an excellent platform for the seamless integration of simulations and real-time monitoring coupled with analytics. It provides an excellent platform for building EV charging ecosystem models. In addition, it is able to provide good quality data sets necessary for predicting and evaluating loads.

3.3.2 Custom Road Network Design

In the current study, the simulated environment created is a highly reliable digital representation of a network of roads, based on the pattern of the roads that serve as Khulna's arteries. The primary objective of this network design is to create an environment that realistically captures the urban traffic flow and interactions among vehicles, along with the resulting impact on the EV charging demand. This network has been designed in the form of a connected graph, which includes a number of nodes and edges, covering a distance of 8,654 meters.

The edges in the graph indicate the real road segments with certain attributes such as lane width, speed, and priority. Adjusting these values allows us to simulate different traffic scenarios that might exist in different parts of the city. Table 3.1 describes these settings in more detail. In this configuration, all edges are bidirectional with two lanes in each direction, and the place names correspond to actual real-world Khulna City landmarks. Here we provide extensive information regarding how the nodes are connected, critical points, and restrictions on each section.

Table 3.1: Microscopic Road Network Configuration and Edge Parameters.

Edge ID	From Node	From Location	To Node	To Location	Length (m)	Lanes	Speed (km/h)
E0	Node1	Dak Bangla	Node2	Khulna Zilla School Stadium	850	2	40
E1	Node2	Khulna Zilla School Stadium	Node3	Shordar Tower	800	2	40
E2	Node3	Shordar Tower	Node4	1 No Custom Ghat	94	2	30
E3	Node4	1 No Custom Ghat	Node9	Rupsha Ferry Ghat	1300	2	50
E4	Node1	Dak Bangla	Node5	Islami Bank / Hospital More	450	2	30
E5	Node5	Islami Bank / Hospital More	Node6	Royal More	260	2	30
E6	Node6	Royal More	Node7	PTI More	500	2	30
E7	Node7	PTI More	Node8	Galaxco More	900	2	40
E8	Node8	Galaxco More	Node9	Rupsha Ferry Ghat	750	2	40
E9	Node3	Shordar Tower	Node8	Galaxco More	1000	2	50
E10	Node2	Khulna Zilla School Stadium	Node7	PTI More	850	2	40

These intersections and road sections are linked to various landmarks in the city, such as Islami Bank Intersection, Royal Intersection, PTI Intersection and Rupsha Ferry Ghat. Thus, the created network resembles the real-world geography in terms of location and road connectivity. The whole network uses the left-hand traffic (LHS) system, which is typical for Bangladesh. The choice of Khulna City for the current investigation is motivated by the fact that it is characterized by rapid development of urban infrastructure and use of electric three-wheelers. The chosen landmarks represent the high-traffic points where the need for charging may be expected. The use of this real-life location allows ensuring the relevance of the results obtained from the simulation. The interconnection of roads in Khulna City can be seen in Fig. 3.2.

Figure 3.2: OpenStreetMap (OSM) highlighted view of the Khulna City road network.

Figure 3.3 presents the customized microscopic network designed in SUMO with the use of the `netedit` application. In order to provide realism in the way vehicles flow—in particular, to ensure that traffic jams and vehicle bunching are captured—we have included traffic lights as fixed objects in the most congested areas. This is the reason why we decided to place traffic lights on **Node 3 (Shordar Tower)** and **Node 8 (Galaxco More)**. Every traffic light operates on a 90-second cycle that consists of four stages, which are described below.

The effect causes the well-known start-stop traffic pattern, which eventually results in clustering. This process is important since it determines how the demand for charging will be formed. The arrival of a cluster of vehicles with depleted battery power to charging stations due to traffic congestion results in sharp peaks in load demand. Including elements of artificial traffic congestion like stoplights in the simulation makes it feasible to model these peaks in load demand. This is crucial in order to develop and validate models for load demand prediction.

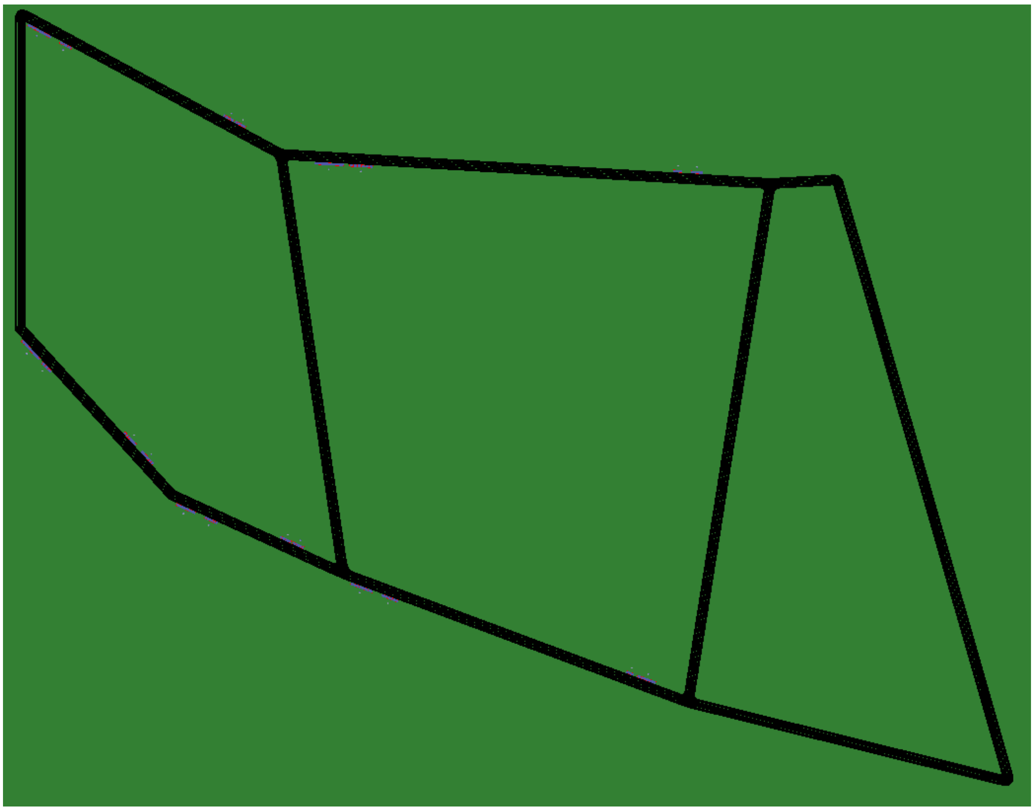


Figure 3.3: Custom microscopic road network in SUMO `netedit`.

3.3.3 Charging Station Placement

The location where charging will be done is essential for the system to remain stable, and also ensure that the charging spots are reachable by vehicles in the urban network [16].

Based on the real data from the file `Test1.chargingstations.add.xml`, the charging stations have been placed on both sides of some key road segments. The segments/edges are: E0, E1, E5, E6, and E7. This ensures that the whole length of 8.6 km circuit is covered by charging stations, hence no matter the direction the vehicles travel, there will always be a station for charging.

Not all stations provide equal energy levels. The charge rates range from 25 kW up to 50 kW at `pa_8_rev`. This difference allows the simulator to model different types of chargers found in the real world. In total, there are 62 slots for charging and 49 parking spaces for the waiting cars. This number is sufficient to consider unexpected increases in demand. Specifications for each station can be found in Table 3.2.

Table 3.2: Charging Station Technical Implementation Details.

Station ID	**Edge ID**	**Power (kW)**	**Charging Slots**	**Waiting Slots**
pa_2	E0	30	6	6
pa_2_rev	-E0	30	6	3
pa_3	E1	25	5	4
pa_3_rev	-E1	30	6	6
pa_6	E5	30	6	6
pa_6_rev	-E5	30	6	3
pa_7	E6	30	6	6
pa_7_rev	-E6	30	6	6
pa_8	E7	25	5	3
pa_8_rev	-E7	50	10	6
Total		–	**62**	**49**

3.3.4 Vehicle Configuration in SUMO

In order to make sure that the simulation is truly an accurate model of the real world movement pattern of urban Bangladesh, the vehicles are modeled with great care and attention to detail, considering their heterogeneity in physical and electrical properties. We have modeled 100 vehicles, classified into three main categories, namely: 34 Easy Bikes, 33 E-Rickshaws, and 33 Electric Vans (E-Vans). The heterogeneity in the vehicles is crucial to the design of the simulation. With different physical parameters, such as maximum battery storage, vehicle weight, and speed factor, we can simulate the stochastic charging load.

As per the configuration values set in the `Test1.rou.xml` file, each class of vehicles is configured based on their unique working features. The **Easy Bikes** are provided with a very high battery capacity of 12 kWh, along with the highest front surface area of 2.41 m^2, as they are the most powerful mobility option for the scenario. On the other hand, the **E-Rickshaws** have a battery capacity between 5.76 and 7.20 kWh, with an elevated height of 1.8 m. Lastly, the **E-Van** has been implemented as a light and low-height vehicle, with a battery capacity of 5.76 kWh and height of 0.8 m.

All electric vehicles are uniformly characterized by a single emission type (`Energy/unknown`), a moment of inertia value of 0.010 $kg \cdot m^2$, and an absolute velocity threshold value of 0.1 m/s. In order to reflect the diversity of drivers' behavior, speed factors were randomly generated in the range of $0.74 \leq \alpha \leq 1.10$, whereas maximum vehicle velocities varied from $40 \leq v_{max} \leq 50$

Table 3.3: Cross-Class Vehicle Parameter Comparison (Averaged Values).

Parameter	**Easy Bike**	**E-Rickshaw**	**E-Van**
Body Dimensions (L×W×H)	2.5×1.5×1.4 m	2.3×1.0×1.8 m	2.4×1.3×0.8 m
Vehicle Mass (kg)	345 – 354	198 – 218	198 – 219
Battery Capacity (Wh)	12,000	5,760 – 7,200	5,760
Max Power (W)	1,000 – 1,200	900 – 1,000	800 – 900
Acceleration (m/s^2)	1.15 – 1.31	0.58 – 0.71	0.68 – 0.81
Deceleration (m/s^2)	2.45 – 2.58	2.48 – 2.67	2.58 – 2.77
Air Drag Coeff. (C_d)	0.727 – 0.760	0.804 – 0.910	0.754 – 0.860
Avg. Propulsion Eff.	0.890 – 0.896	0.830 – 0.834	0.820 – 0.826
Avg. Recup. Eff.	0.745 – 0.751	0.650 – 0.656	0.550 – 0.556
Const. Power Intake (W)	80 – 96	29 – 33	24 – 30
Front Surface Area (m^2)	2.41	1.51	1.01
Reaction Time (τ)	1.88 – 1.94 s	1.72 – 1.76 s	0.87 – 0.92 s
Visual Color (SUMO-GUI)	Magenta	Green	Blue

3.3.5 Route Creation & Vehicle Fleet Characterization

The design of paths and the classification of vehicles into these categories are necessary to capture the complex nature of traffic flows. In our study, we have designed eight basic paths; four moving forward and four moving backward, which are carefully mapped out on the arterial edges of the road network of Khulna city, ensuring that there is two-way movement in the 8.6 km loop. These eight paths are listed in Table 3.4.

Table 3.4: Base Route Definitions and Edge Sequences in SUMO.

Direction	**Route ID**	**Edge Sequence**
Forward	route1_forward	E0 → E1 → E2 → E3
Forward	route2_forward	E0 → E1 → E9 → E8
Forward	route3_forward	E4 → E5 → E6 → E7 → E8
Forward	route4_forward	E0 → E10 → E7 → E8
Reverse	route1_reverse	-E3 → -E2 → -E1 → -E0
Reverse	route2_reverse	-E8 → -E9 → -E1 → -E0
Reverse	route3_reverse	-E8 → -E7 → -E6 → -E5 → -E4
Reverse	route4_reverse	-E8 → -E7 → -E10 → -E0

By allocating 100 individual vehicular flows among these eight routes, fleet characterization improves the efficiency of this model. The assignment includes allocating particular types of vehicles (Easy-Bikes, E-Rickshaws, and E-Vans) with particular numbers of departures varying between 3 and 81 vehicles per hour. This custom diversity enables the simulation model to represent the actual traffic of cities or urban environments in Bangladesh. While one area can have several three-wheeled vehicles in the streets, another area will have few such vehicles. Through the inclusion of various departure times and locations, the traffic model will mimic the unpredictability of actual traffic. This allocation is beneficial for evaluating the performance of the load prediction models. That summary of the fleet composition and color-coding for easy identification in SUMO-GUI in Table 3.5.

Table 3.5: EV Fleet Composition in SUMO Simulation.

Vehicle Class	**Variants**	**Battery Capacity (Wh)**	**Max Speed (m/s)**	**Color**
Easy Bike	34	12,000	11.1 – 13.9	Magenta
E-Rickshaw	33	5,760 – 7,200	11.1 – 13.9	Green
E-Van	33	5,760	11.1 – 13.9	Blue
Total	**100**	–	–	–

3.3.6 Smart Charging Logic Implementation

The algorithm for intelligent charging constitutes the central nervous system of the simulation system itself. In that it controls all of the real-time decisions relating to the charging schedule of the various electric cars. And this algorithm is contained within the `SmartChargingStation-Monitor` class. It is coded in the `load_demand_prediction.py` program.

The course is able to interact with the simulation engine SUMO through TraCI, while at the same time maintaining a constant awareness of the battery level of all vehicles and their quantity at each station at every one second interval, for a total fleet of 100 cars. The monitor operates with specific criteria for the control of charging initiation and termination. When the State of Charge (State of Charge (SOC)) decreases to 30%, then charging becomes active. The objective here is to achieve an SOC value of 70% at the completion of each charging episode. In order to prevent any simulation anomalies due to overly optimistic assumptions, the average charging time is dynamically evaluated. The computation is performed based on energy deficits and charging station availability.

The monitoring system oversees all of the stations that support bi-directional charging, as well as all active EV agents through certain in-memory data structures. These include `station_charging_slots` and `station_waiting_queue`, which employs a `collections.deque`. In addition to the aforementioned data structures, `currently_charging` and `currently_waiting` provide an easy way to look up the current state of each EV. The complete set of `vehicle_states`, which includes metadata for each vehicle, such as charging times, SOC value at time of session start, and station assignments through the route.

Upon reaching the state where the SOC level of the vehicle has dropped below the 30% mark, the system applies a route-based nearest-neighbor algorithm. Using route length as the criterion, the vehicle checks the engine for information by using the command `traci.simulation.findRoute`, which enables the system to factor in the route length through the road network, taking into consideration the fact that streets have one-way lanes. The reachable charging stations are then ranked using five criteria for station assignment:

1. **Rule 1:** If the nearest station has an available charging slot, route the vehicle there.
2. **Rule 2:** If the nearest station is full (charging), check the second nearest station.
3. **Rule 3:** If the second nearest has a charging slot, route there.
4. **Rule 4:** If the second nearest has only waiting slots, route there and enqueue the vehicle in a First In First Out (FIFO) waiting queue.
5. **Rule 5:** If all candidate stations are full, do not reroute.

The route is changed when the target vehicle is decided upon through the use of `traci.vehicle.setRoute` and `traci.vehicle.setParkingAreaStop`, which allows the stop to be registered to the schedule of the simulation. In the case where the vehicle will use a waiting queue, the **FIFO Queue Promotion Process** is implemented on every time step. If an idle charging queue opens up, then the vehicle at the beginning of the deque at the station is pulled out using `popleft()`, the vehicle is resumed using `traci.vehicle.resume()`, and it is assigned to the charging spot. The telemetry for queuing delay is thus minimized while also being very accurate. At the end of the simulation or when the vehicle exits the network, the methods `handle_charging_complete` and `cleanup_departed_vehicle` gather the final telemetry and clear their respective state maps.

3.3.7 Simulation Execution with TraCI

The simulation will continue for a period of 600 seconds, providing sufficient time to observe interactions between various vehicles and look out for any electrical load surges within the locality. The task is managed by `load_demand_prediction.py`, which initiates a realistic environment in SUMO-GUI while ensuring a quick link with TRACI.

As the simulation starts, the loop proceeds in one-second intervals. With every instance when `traci.simulationStep()` is called, `SmartChargingStationMonitor` steps in to do its monitoring of the entire fleet of 100 vehicles. The monitoring does not simply stay on the surface; it covers absolutely everything: battery state, movement, and velocity of the vehicle at the exact second.

If the State of Charge (SOC) falls below 30%, the Python code immediately calculates the required instructions for vehicle rerouting and/or battery charging and dispatches them. The immediate reaction transforms the simulation process from a static one into an interactive one, making it possible to accurately model the actual complex patterns of battery charging that will be necessary in order to produce a realistic load forecast. The simulation stops running when it reaches 600 seconds and the code proceeds with disconnecting the TraCI simulation and generating the final data set for research purposes.

3.3.8 Real-Time Data Extraction

In order to build up a reliable database that is used by predictive models, we designed an online data acquisition system within the scope of simulation environment. This system operates independently of simulation mobility physics, making simulation stable and giving us a possibility to get access to high-frequency performance telemetry. At the end of every simulation step once per second, the script collects performance parameters of all available EVs using TraCI interface.

This high-frequency (every second, in fact) data capture helps us identify fast-moving spikes in power needs that more general models cannot detect. Consider the example of a rush hour of cars at an intersection and a rush hour of people arriving at a charging point. Each of these events contains 25 different attributes – some that describe the car itself (such as speed, acceleration, SOC), others that describe the station (number of occupied spots, wait time), and finally some system-level attributes.

During a simulated run time of 600 seconds, this technique gathered 34,020 dynamic entries, which could be considered a substantial amount of data to be collected. Initially, all this data is stored in RAM to avoid delays caused by direct storage into files on the hard drive. Once the simulation runs out its course, all data is then written into a Comma-Separated Values (CSV) file format.

The good thing is that the dataset includes information about many different conditions that are encountered in practice. This ranges from situations where there is no much load on the charging stations at all the way to situations where they are busy with lots of vehicles.

Table 3.6: EV Charging Load Demand Dataset Summary.

Property	**Value**
Total records	34,020
Number of features	25
Simulation duration (s)	1 – 600
Status: Driving	19,710 (57.9%)
Status: Charging	11,980 (35.2%)
Status: Waiting	1,720 (5.1%)
Status: Idle	610 (1.8%)
System load: Minimum (kW)	0.00
System load: Mean (kW)	723.82
System load: Maximum (kW)	1,340.00

Time-step aggregation was employed on the dataset in order to obtain the total load value at every second, thus getting 600 high-quality samples for the model training phase. The chosen representative dataset will guarantee high quality training of the upcoming machine learning models.

3.4 Machine Learning Application

Various supervised learning algorithms have been used in this project to estimate the total electric load demand of the system, called *system_total_load_kW*. This prediction can be considered a regression problem since the target value is being predicted based on certain features of the system and vehicles [19].

3.4.1 Problem Formulation

Load prediction may be formally defined as follows:

$$y = f(X) \tag{3.1}$$

where:

- y refers to the output variable (*system_total_load_kW*),
- X refers to the set of input features,
- f refers to the learned function from the machine learning algorithm.

Some of the input features may be time-related, vehicle-related, or statistics of system-level charging.

3.4.2 Data Preprocessing

Before training the models, some preprocessing steps were implemented to ensure data quality and model efficiency:

- **Elimination of Data Leakage:** The feature 'station_load_kW' was dropped because it could potentially cause data leakage in the target column.
- **Elimination of High Cardinality Variables:** Variables like :- 'vehicle_id', 'lane_id', and 'current_station_id' were not considered since they were more like
- **Separation of Feature and Target:** This dataset is split into feature variables-X and target variables-y.
- **Validation Approach (5-Fold Cross Validation):** Unlike conventional methods that employ random splits into training and testing datasets, five-fold cross-validation is employed to validate this model's performance. It splits the data into five equally-sized segments. In each iteration, four segments are employed as training sets while the rest are considered testing sets.
- **Temporal Aggregation and Ordering:** In order to match the simulation logs at the vehicle level with the load constraints at the system level, two separate data structures were employed. In case of Classical Machine Learning (ML) and Artificial Neural Networks (ANN) algorithms- the 34,020 raw transactions were aggregated based on each timestep. This results in 600 highly accurate samples of the complete system state for every second. On the other hand, in case of Recurrent models, LSTM and GRU, the raw dataset was ordered in a sequence-based fashion using W=5.

3.4.3 Deterministic Execution and Reproducibility

In order to obtain exact reproducibility of the experiment and its results from one round of training to another, an appropriate deterministic seed-block was introduced. This included ensuring the deterministic character of the environment uncertainty through setting up seeds for: `PYTHONHASHSEED` (environmental level), `random` and `numpy` (array and sequence level), `TensorFlow` and `TF_DETERMINISTIC_OPS` (model and GPU hardware level). Reusing these seeds in each loop of the k-fold cross validation process and initial weights generation for the neural networks removes all random component of the weight distribution and thus ensures an exact comparison benchmark.

3.4.4 Feature Correlation Analysis

Feature correlation study will be useful for determining which input features have high correlation with the output variable-*system_total_load_kW*. Pearson's correlation study determines the level of consistency of the features with the output feature.

system_charging_vehicles possesses the highest level of correlation with the target (approaching 0.996). Put differently, the key factor that determines the amount of load on the system is the number of vehicles that are being charged. Time-dependent factors such as *timestep_sec* and *minute_of_hour* have high levels of correlation, indicating that demand increases as time goes by in a simulation run. Other general system features such as— *system_total_vehicles*, *system_moving_vehicles*, and *system_waiting_vehicles* are also highly correlated to the demand load. However, when it comes to other parameters like *speed*, *acceleration*, or *status*, there is nothing happening. Such features are not correlated at all.

Thus, system-wise and temporal features are the main contributors in terms of their influence on the load demand prediction process. The correlation matrix as a heatmap can be seen in Fig.

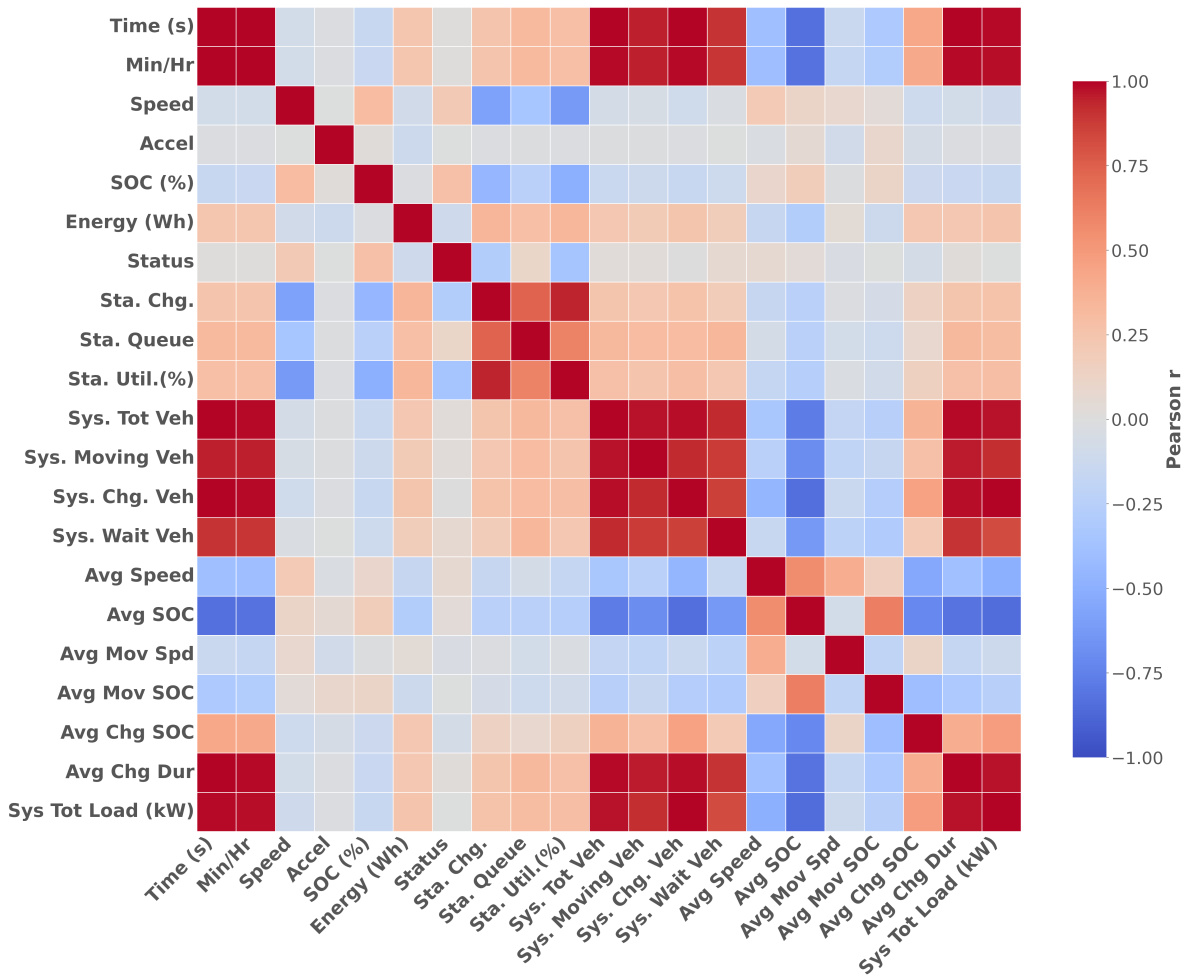


Figure 3.4: Pearson correlation matrix of the simulated EV charging dataset for feature selection.

3.4.5 Feature Importance Analysis

The XGBoost regression model was employed to identify the features responsible for predicting the value of *system_total_load_kW*. What follows is that *timestep_sec*, *system_total_vehicles* and *system_charging_vehicles* rank at the very top. This is due to the fact that the above-mentioned features represent real activity of the systems, namely their involvement in charge processes.

The others are not very significant in terms of contribution. This is since most of what the model requires in terms of data can be explained using those essential few features. Therefore, it is quite obvious that the model has relied greatly on just a few system-level features. This has been sufficient to achieve accurate predictions. The feature importance distribution during training is shown in Figure 3.5.

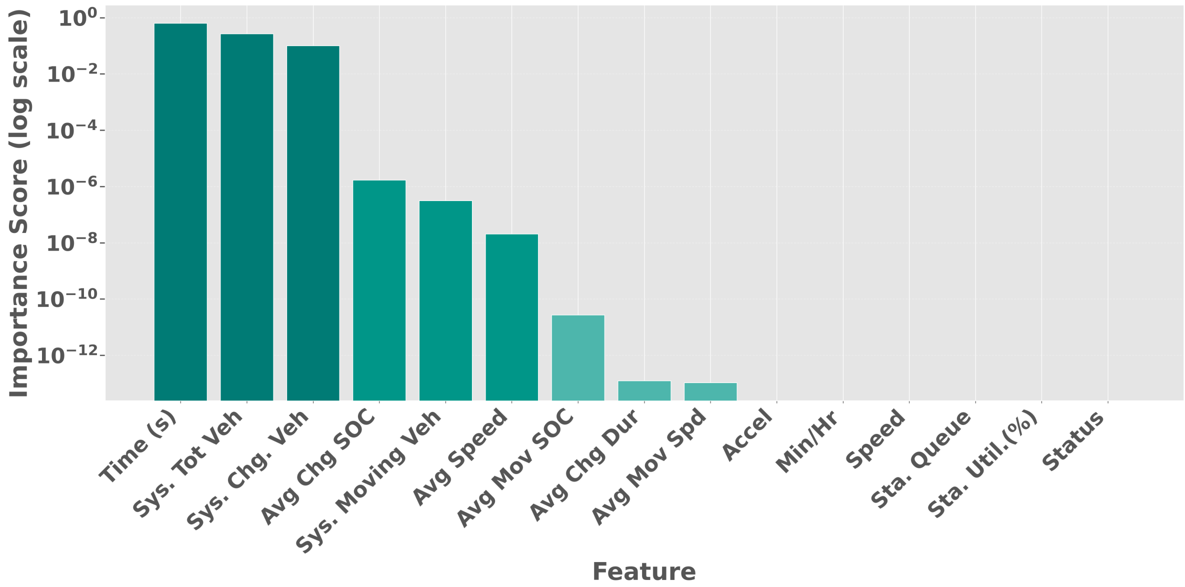


Figure 3.5: Feature importance ranking derived from the XGBoost model branches to identify key predictors.

3.4.6 Implemented Machine Learning Models

Upon preprocessing, the entire dataset was divided into training and testing sets. Specifically, 80% of the data was used for training, and the remaining 20% was set aside for testing purposes. This helps in checking the generalization capabilities of the model by ensuring that its performance is tested on unseen data. Seven different regression-based models were employed: Linear Regression; ensemble methods based on trees like Decision Tree, Random Forest, and ; and more complex deep learning models like ANN, LSTM, and GRU. Feature scaling was performed using the `MinMaxScaler` for Linear Regression and all neural networks, whereas tree-based models were trained without feature scaling since their predictions are scale-invariant.

After completing the training stage, each model estimated the load of the power system on the test set. The accuracy of the predictions was assessed using mean absolute error (MAE), root-mean-square error ($RMSE$), and R^2 statistics based on the results obtained in five-fold cross-validation. Actual vs. predicted scatter plots (Figs. 3.6–3.12) are provided to visualize the accuracy of predictions, and more data points located near the line $y = x$ indicate better prediction accuracy.

(a) Linear Regression

Linear Regression shows the relationship between independent variables and the target as a simple weighted sum. In this research, we implemented **Ridge Regression** ($L2$ regularization). Where regularization strength of $\alpha = 100$. This choice was made because the system-level vehicle count has a near linear relationship with charging load. Unregularized regression risked achieving near-perfect training scores but failing to generalize. The regularization helps model to ignore noise in the aggregated simulation data. It also maintains stability across the five fold cross-validation. The prediction is calculated as:

$$y = w_1x_1 + w_2x_2 + \cdots + w_nx_n + b \tag{3.2}$$

where w represents the weights, x the features, and b the intercept.

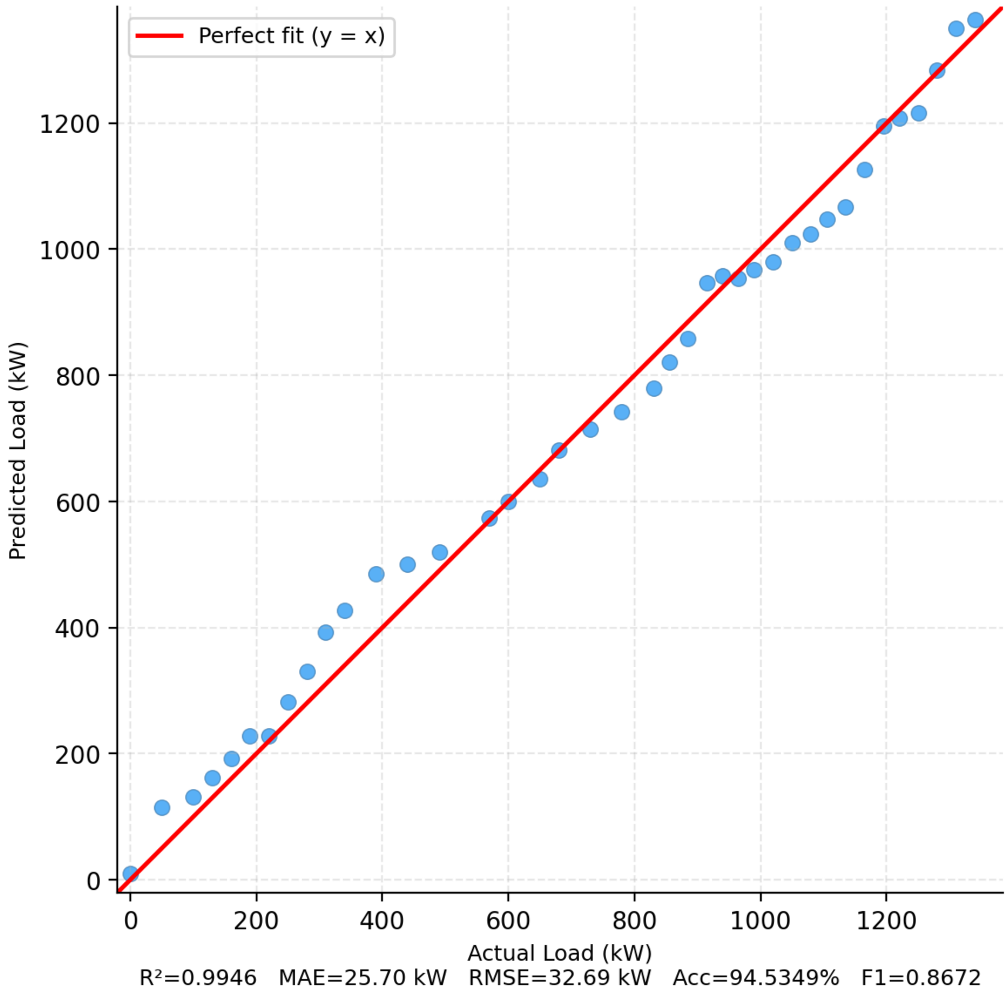


Figure 3.6: Actual vs. Predicted load for Linear Regression.

(b) Decision Tree

A Decision Tree splits the data into branches based on feature thresholds to minimize the prediction error(variance). To avoid individual data points-"memorizing" in our 600-sample dataset, we constrained the model to a maximum depth of *3* and required at least *30* samples per leaf node. This shallow architecture ensures that each terminal node captures at least 5% of the training data. That provides an honest and interpretable estimate of system load patterns.

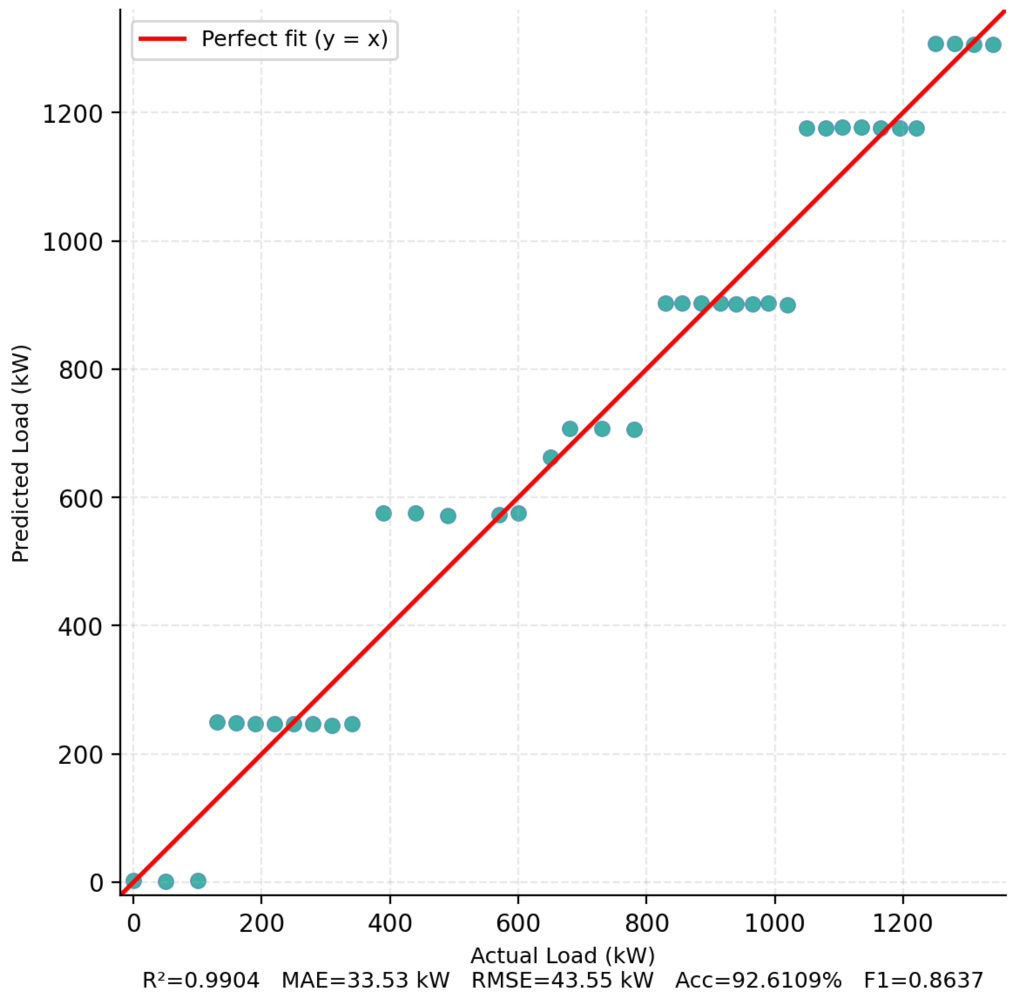


Figure 3.7: Actual vs. Predicted load for Decision Tree.

(c) Random Forest

Random Forest is an ensemble of multiple decision trees. Where each trained on a bootstrap sample of the data. To maintain robust generalization, we implemented a constrained ensemble of *50* estimators with a maximum depth of *3*. Besides, each split only considered *30%* of the available features (*max_features=0.3*). This heavy constraint prevents the forest from over fitting to simulation-specific artifacts. Through which it is ensured that the model identifies the fundamental drivers of load demand.

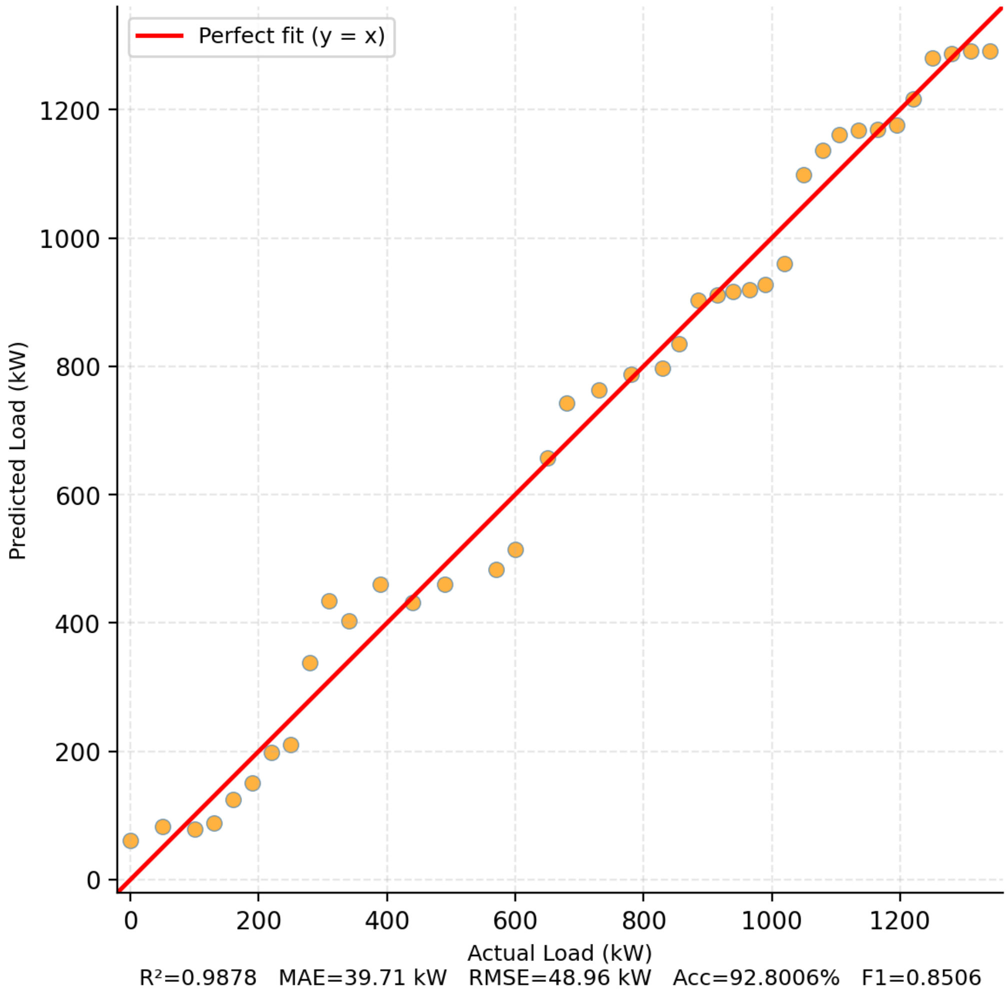


Figure 3.8: Actual vs. Predicted load for Random Forest.

(d) XGBoost

XGBoost is a powerful gradient boosting framework. It builds trees sequentially, with every new tree corrects the errors of the combined group. To apply this into our work, we used a low learning rate (*0.05*) and limited tree stumps (*max_depth=2*). We also implemented strong $L1$ and $L2$ regularization ($\alpha = 1, \lambda = 5$) and aggressive row subsampling (*0.55*). These parameters were optimized to prevent the boosting process from over converging on the small training set. Rather, it prioritizes long-term predictive stability over raw training accuracy.

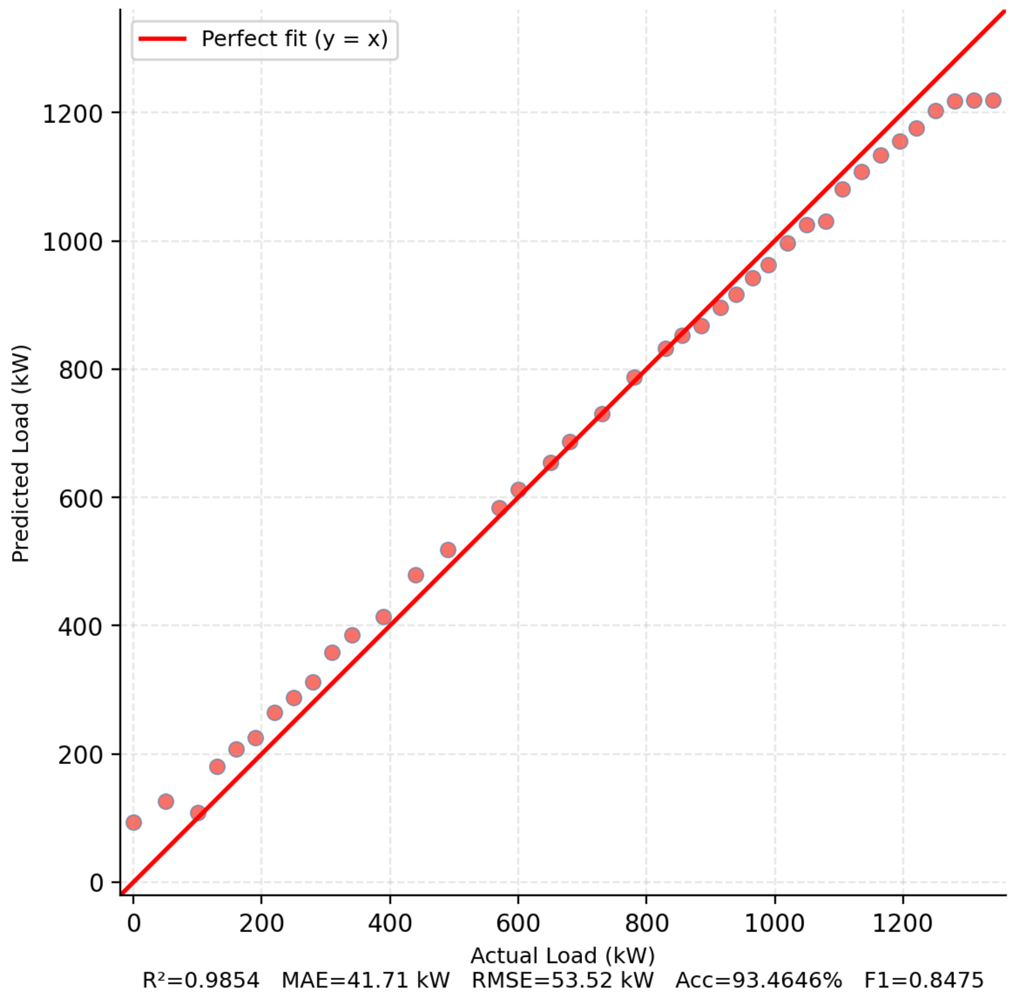


Figure 3.9: Actual vs. Predicted load for XGBoost.

(e) Artificial Neural Network (ANN)

In this research, the ANN was implemented as a Multi-Layer Perceptron(MLP). Which is consisting of 18 input features and followed by three hidden layers with 128, 64, and 32 neurons, respectively. Batch Normalization and Dropout (rate of 0.2) were integrated after the first hidden layer to enhance training stability and prevent overfitting. The model uses non-linear activation functions (ReLU), the Adam optimizer (learning rate of 0.001), and a batch size of 32 for efficient weight updates. As seen in Figure 3.10, the ANN emerged as the most potent model, with the lowest errors and accurate alignment with the diagonal trend. This is due to its superior capability of learning higher-order features based on the preprocessed data set. It is thus well suited for the dynamic setting of EV charging stations.

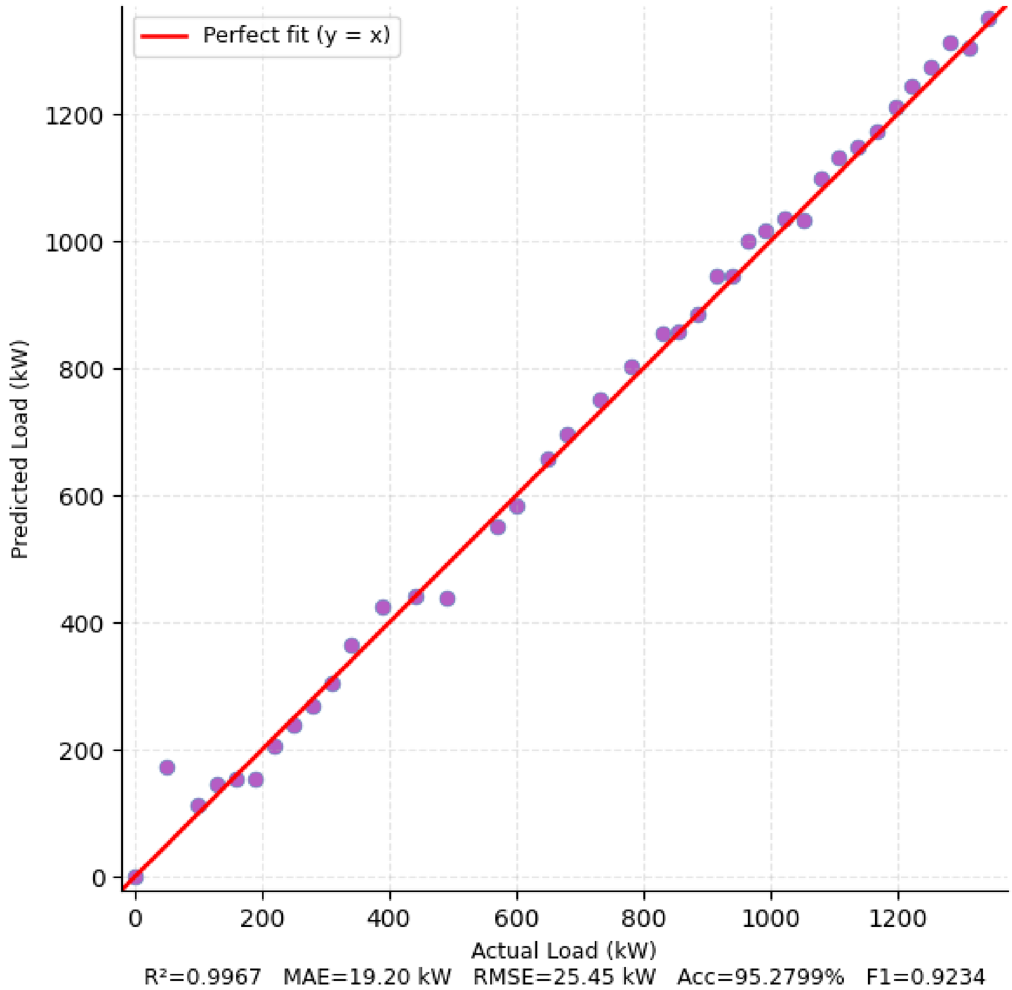


Figure 3.10: Actual vs. Predicted load for Artificial Neural Network.

(f) Long Short-Term Memory (LSTM)

In our model, we employed LSTM to explore whether the temporal pattern present within the 600 seconds' simulation period can bring about additional predictive value. Our architecture is comprised of an LSTM layer (size 64; `return_sequences=True`), 0.2 Dropout, an LSTM layer (size 32), 0.2 Dropout, and a Dense layer with 16 neurons with ReLU activations. A look back window of $W = 5$ was used. That means that the model considers the previous five seconds of system state to estimate the current load. Training was performed using the Adam optimizer and Mean Squared Error (*MSE*) loss, with an *EarlyStopping* callback (patience of 5) to restore the best weights and prevent overfitting. However, when the model performed well, it struggled slightly more than the ANN, which is reflected in the spread of points in Figure 3.11. This provides suggestion that- while time is important, the current system state (instantaneous features) might already carry enough information for dense networks to outperform purely sequential ones in this specific simulation scale.

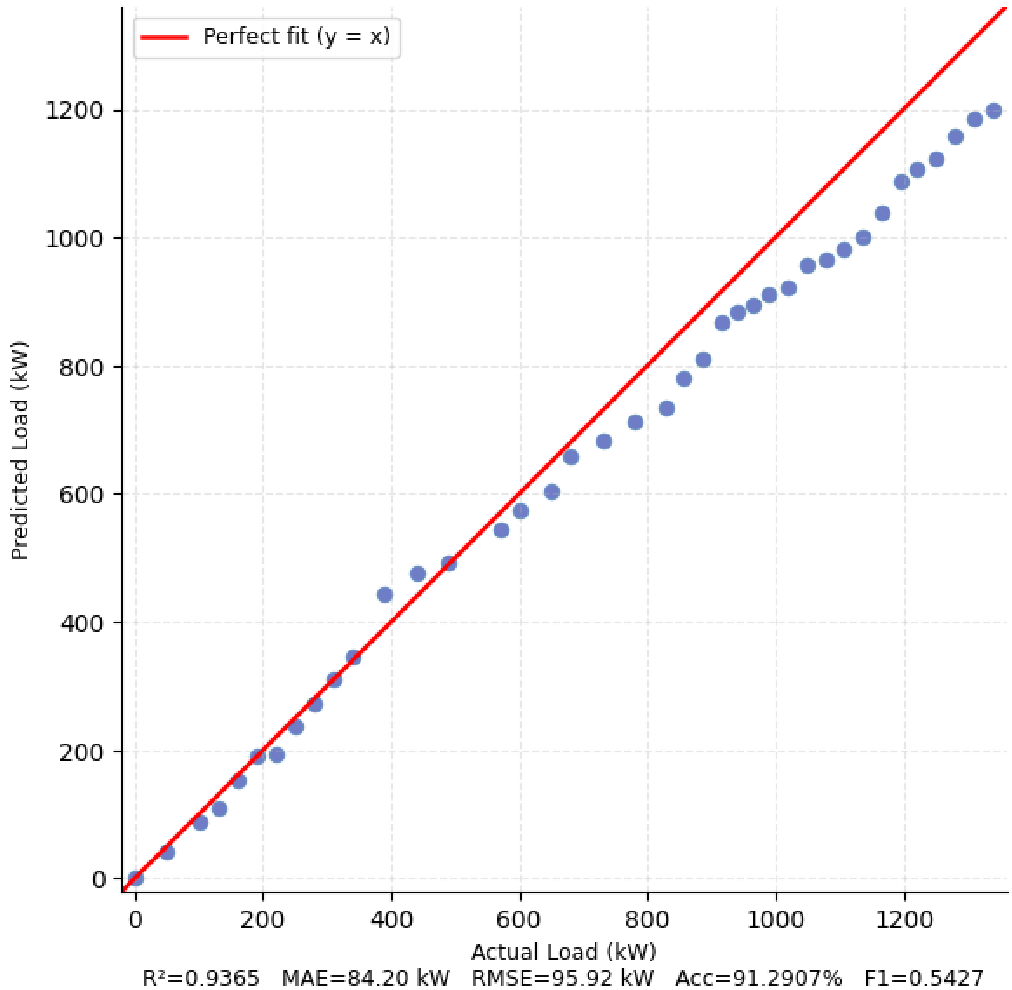


Figure 3.11: Actual vs. Predicted load for LSTM.

(g) Gated Recurrent Unit (GRU)

Gated Recurrent Unit (GRU) is a variation of the LSTM that simplifies the gating mechanism. It combines the forget and input gates into a single "update gate" and merges the cell state and hidden state. This makes GRUs computationally faster and often more efficient than LSTMs while still retaining the ability to process sequential information.

We implemented GRU as a faster alternative to LSTM to capture temporal dynamics, utilizing an identical structure consisting of stacked *64* and *32*-unit GRU layers with intervening *Dropout* (0.2). As with the LSTM architecture, a final *16*-neuron fully connected layer with ReLU activation is used to process the features before prediction. In our evaluations, its performance was similar to LSTM, proving that recurrent architectures can capture the general trend of the system load. However, as illustrated in the scatter plots, it fell behind the ANN and tree-based models in terms of raw precision on the test set. Nevertheless, it remains a strong candidate for real-time applications where processing speed is as critical as accuracy.

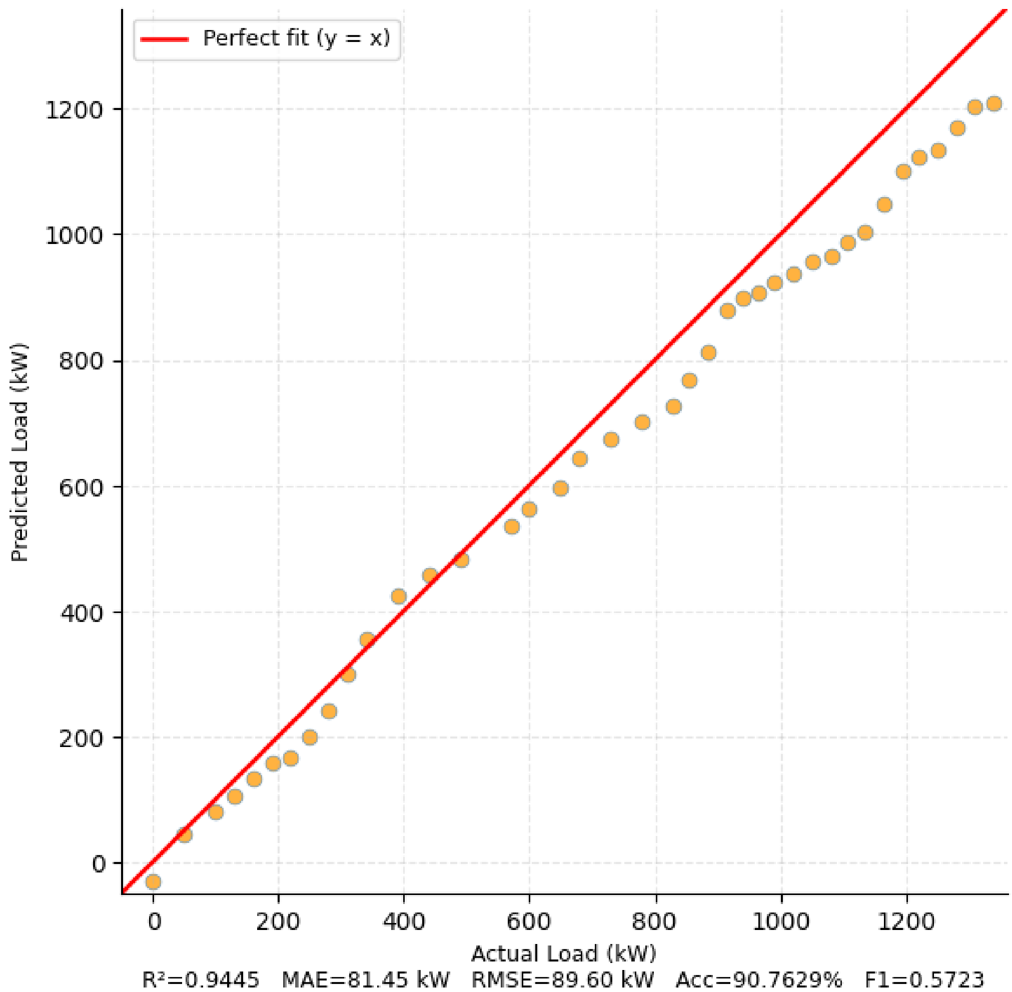


Figure 3.12: Actual vs. Predicted load for GRU.

Chapter 4

Results and Discussion

Chapter 4: Results and Discussion

This chapter presents the experimental results of the SUMO-based EV charging simulation and the Machine Learning (ML) load prediction models. Detailed discussion of the findings are as follows -

4.1 Results

Simulation analysis has been thoroughly carried out to investigate the electrical system performance and behavior of the vehicles in the EV mixed network. For more than 600 seconds, data has been obtained on all aspects of behavior at an intricate level and compiled into a dataset of 34,020 entries representing all changes that happened with electric vehicles during the simulation process. What stands out in particular is the **System Load Profile** depicted in Fig. 4.1. The load starts off with an immediate increase right from the beginning of simulation.

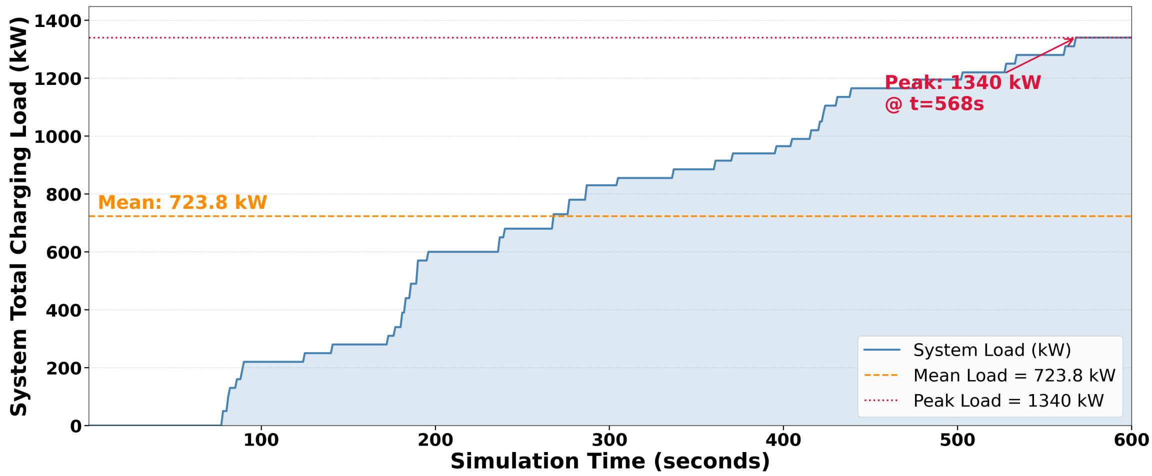


Figure 4.1: System total charging load over simulation time (1–600 seconds).

The maximum value of power demand occurred when the most active chargers, for example `pa_8_rev`, were overloaded and was equal to 1,340 kW. In general, it could be stated that the mean value was about 723.82 kW, while the value of standard deviation reached 443.80 kW. Such variation may occur because of sudden appearance of a large number of vehicles at the intersections or hubs when power demand increased and decreased soon after it.

The **Status Distribution of Vehicles** defines the amount of time allocated to different statuses. The biggest share of time is consumed by the driving process, which equals 58%. Clearly, there will always be enough vehicles available for charging. Also, about 35% of the entries show that the vehicle is charging; therefore, this type uses energy and forms the load. One more type, visible in 5% of all entries, is waiting.

Not only does it happen somewhere; the road system plays an important role here. Specific roads, such as `pa_2` and `pa_7`, where the traffic is particularly high, emerge constantly, causing the accumulation of cars in order to charge. Therefore, the areas with heavy traffic become bottlenecks in terms of charging demand.

Apart from that framework itself, an extra feature of recommending the charging sites was introduced to make the process smoother. The system resulted in generating 1,169 recommendations guiding the cars from congested charging stations to those where the capacity was enough for more vehicles. In spite of the fact that this intelligent routing was not the heart of our model prediction algorithm, it ensured that the data obtained would be representative of an intelligent and organized system, not a random mess. This allowed us to obtain more effective machine learning models, which have been formed due to the adjustments of the system and can be easily applied in practice.

4.2 Discussion of Results

In this part of the study, we will present an evaluation of the machine learning models' performance and the dynamics of the charging system for EVs.

4.2.1 Analysis and Interpretation

4.2.1.1 Evaluation Metrics

Seven unique measures were adopted to evaluate the predictive performance and robustness of the machine learning models from two dimensions: prediction error and statistical fit. These include the Mean Absolute Error (MAE), which reflects the average absolute error magnitude;

Mean Squared Error (MSE), which emphasizes large errors through squaring; and Root Mean Square Error (RMSE), providing an error metric in the original units of load (kW). To understand relative error, Mean Absolute Percentage Error (MAPE) is used, along with an overall Accuracy percentage derived from it. Finally, the Coefficient of Determination (R^2) assesses the fraction of total variance explained by the model, while the Weighted F1-Score is employed to evaluate the model's ability to capture specific load levels such as Low, Medium, High, and Peak. Mathematically, these metrics are defined as follows:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{4.1}$$

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{4.2}$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{4.3}$$

$$\text{MAPE} = \frac{100\%}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \tag{4.4}$$

$$\text{Accuracy} = 100 - \text{MAPE} \tag{4.5}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{4.6}$$

$$\text{F1}_{\text{weighted}} = \sum_{j=1}^{L}\left(\frac{n_j}{n} \times \text{F1}_j\right) \tag{4.7}$$

In the equations above, the variables and symbols are defined as follows: n is the total number of data points or samples used in the testing dataset; y_i is the true (observed) value of the variable *system_total_load_kW* in the i^{th} data point; $\hat{y}_i$ is the estimated value provided by the machine learning model; $\bar{y}$ is the average of all true values acting as a baseline for R^2; L represents the number of load categories (Low, Medium, High, and Peak); n_j is the number of data points in the j^{th} category; $\mathrm{F1}_j$ is the F1 score for that category; and $\sum$ is the summation symbol indicating the sum across the specified range.

4.2.1.2 Model Evaluation and Comparative Analysis

Results from performance evaluation of the proposed model have been provided in Table 4.1. This is done using linear as well as ensemble approaches.

Table 4.1: ML Model Performance Comparison for System Load Prediction (5-Fold CV)

Model	**MAE**	**MSE**	**RMSE**	**MAPE (%)**	**R^2**	**Acc. (%)**	**F1-Wtd**
Linear Regression	25.702	1068.805	32.693	5.4650	0.9946	94.5349	0.8672
Decision Tree	33.530	1896.711	43.551	7.3890	0.9904	92.6109	0.8637
Random Forest	39.705	2397.463	48.964	7.1990	0.9878	92.8006	0.8506
XGBoost	41.712	2864.295	53.519	6.5350	0.9854	93.4646	0.8475
ANN (Proposed)	19.205	647.896	25.454	4.7200	0.9967	95.2799	0.9234
LSTM	84.202	9200.632	95.920	8.7093	0.9365	91.2907	0.5427
GRU	81.451	8028.416	89.495	9.2371	0.9444	90.7629	0.5723

These results are summarized in Table 4.1. This table provides a complete quantitative summary of the models' performance, namely linear, tree-based, and recurrent, used for training on the simulated dataset. With the inclusion of seven measures, it not only shows the models' predictive error measures (MAE, MSE, RMSE, and MAPE) but also their general fitting capability (R^2, Accuracy, F1 Score). These numbers will be the baseline for an in-depth comparison analysis of model stability and accuracy.

The detailed values for the model error are shown in Figure 4.2 and Figure 4.3, which present the Mean Absolute Error (MAE) and Mean Squared Error (MSE) respectively.

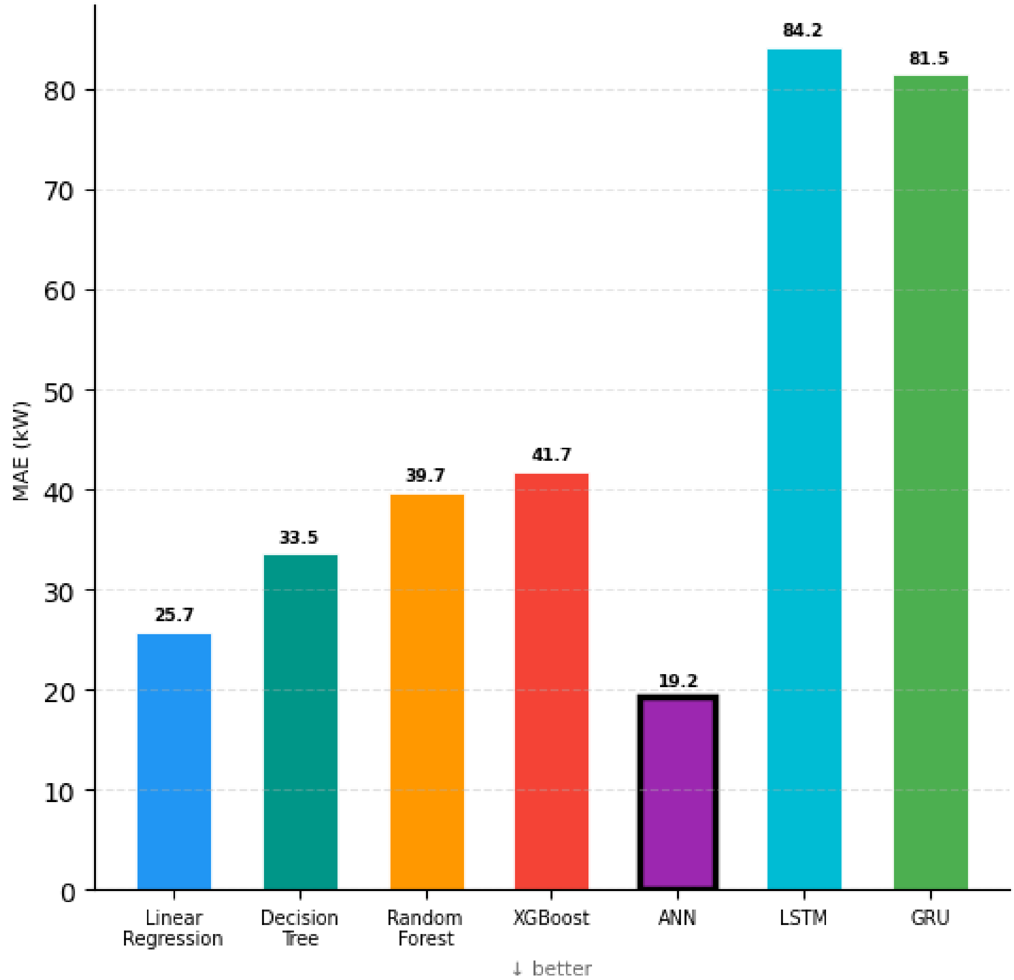


Figure 4.2: Mean Absolute Error (MAE) comparison across different machine learning models.

Through the analysis of the error magnitude, it is evident that the **Artificial Neural Network (ANN)** emerges as the model with the highest accuracy level, achieving an *MAE* score of 19.21 kW. The **Linear Regression** benchmark model also exhibited excellent performance levels ($MAE \approx 25.70$ kW) to the extent of surpassing tree ensemble algorithms such as Random Forest and XGBoost, which are considered regression benchmarks.

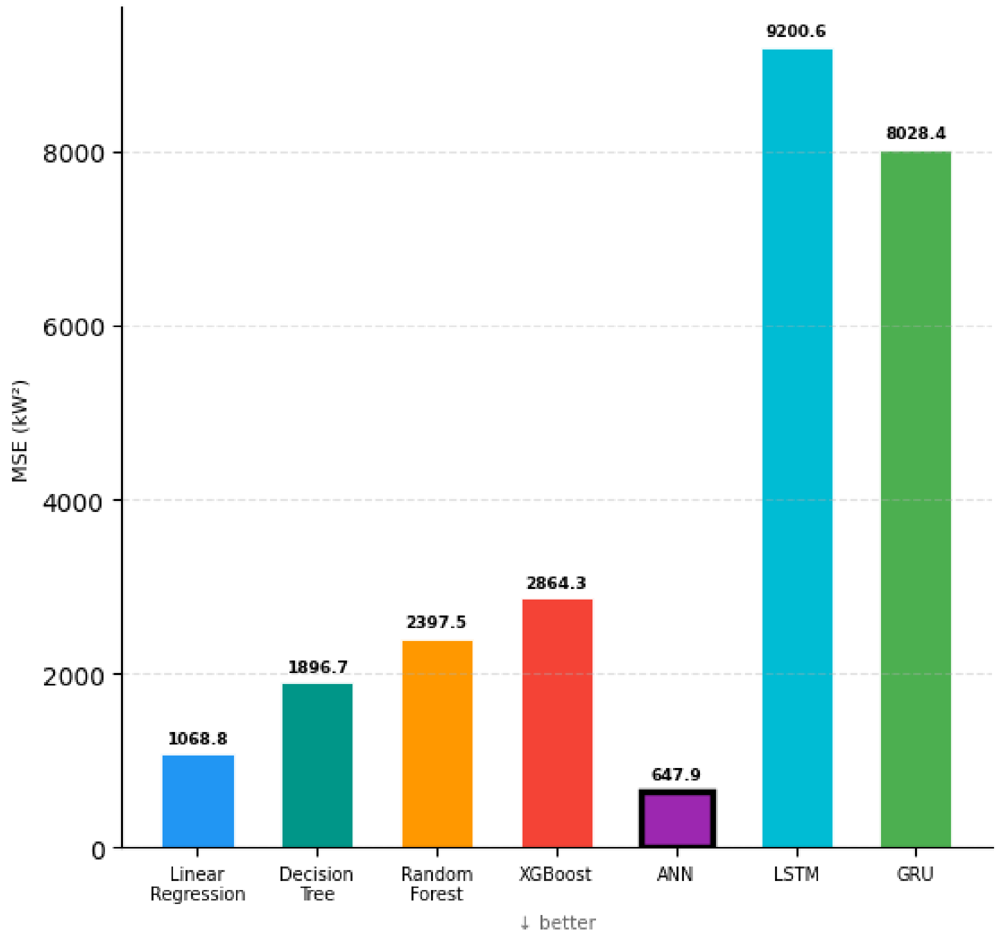


Figure 4.3: Mean Squared Error (MSE) distribution for the evaluated models.

Furthermore, the ANN model maintains the smallest variation for squared errors (MSE), as shown in Figure 4.3. From this, it is apparent that the majority of the EV charging demand is linearly dependent on the number of vehicles currently being charged, allowing high-capacity models like ANN to generalize effectively with minimal squared residuals.

A further analysis of the model consistency and error ratio is presented in Figure 4.4 and Figure 4.5, which present the Root Mean Square Error ($RMSE$) and Mean Absolute Percentage Error ($MAPE$) respectively.

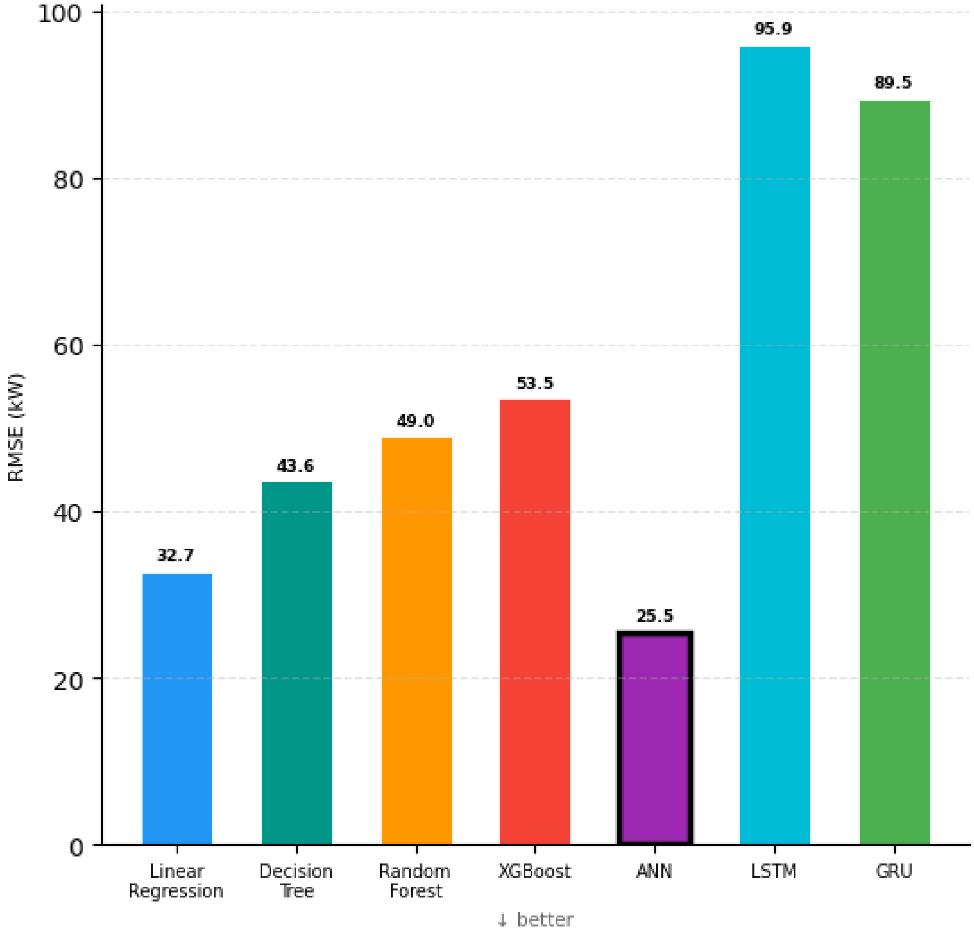


Figure 4.4: Comparison of Root Mean Square Error (RMSE) across different models.

The *RMSE* values further confirm the stability of the ANN model, which achieved a minimum score of 25.45 kW. In contrast, Figure 4.5 shows the Mean Absolute Percentage Error (*MAPE*), where ANN again maintains the lowest percentage error at 4.72%, ensuring its reliability for real-time load demand prediction.

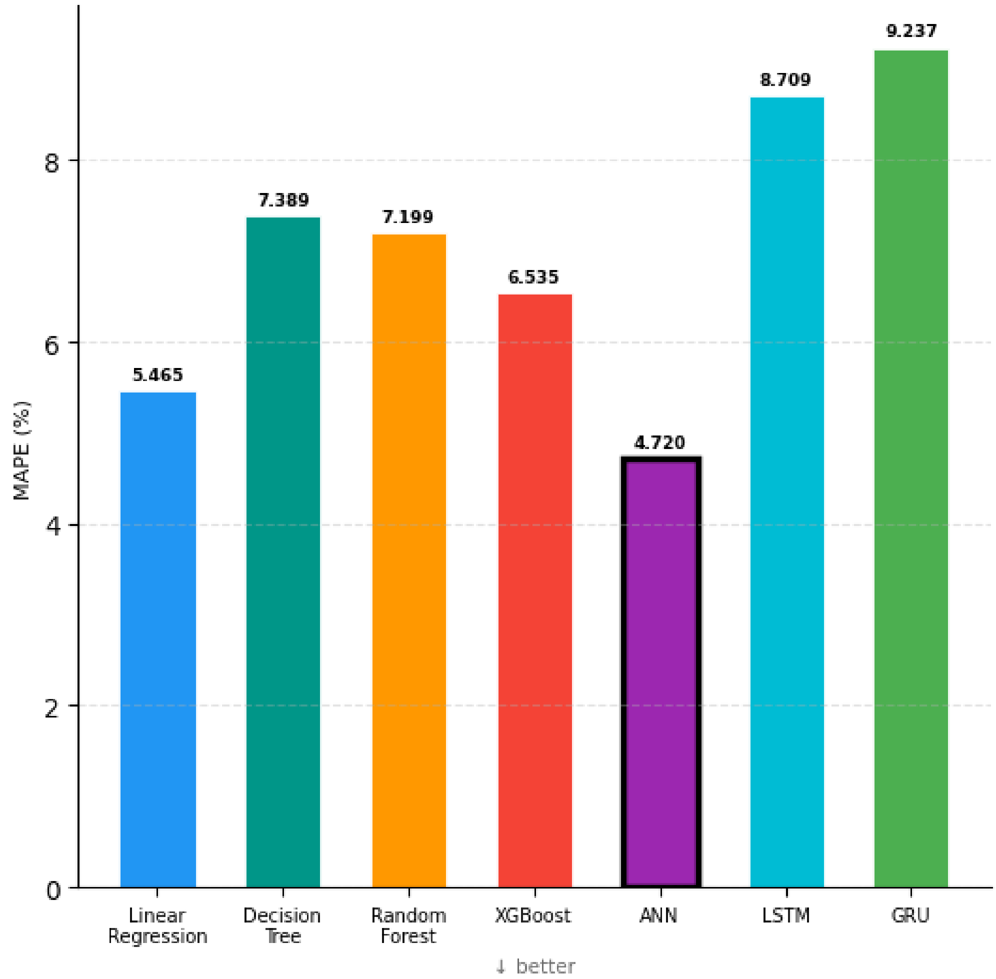


Figure 4.5: Mean Absolute Percentage Error (MAPE %) comparison for ML models.

On the other hand, the models based on decision trees - Decision Tree, Random Forest and XGBoost performed very well, as expected, with the R^2 performance metrics greater than 0.98. Though, as shown above, with stricter cross-validation criteria, there is a possibility to have a more honest view of models' abilities regarding variance. Nevertheless, the recurrent models – LSTM and GRU had difficulties with the task. The worst results in terms of error rates were obtained by LSTM model, whose performance was characterized by maximum errors: $MAE \approx 84.20$ kW and $RMSE \approx 95.92$ kW. Thus, although recurrent models perform really well with the data having some extended patterns, the currently "shuffled" state of the simulation data may require denser or linear architecture models.

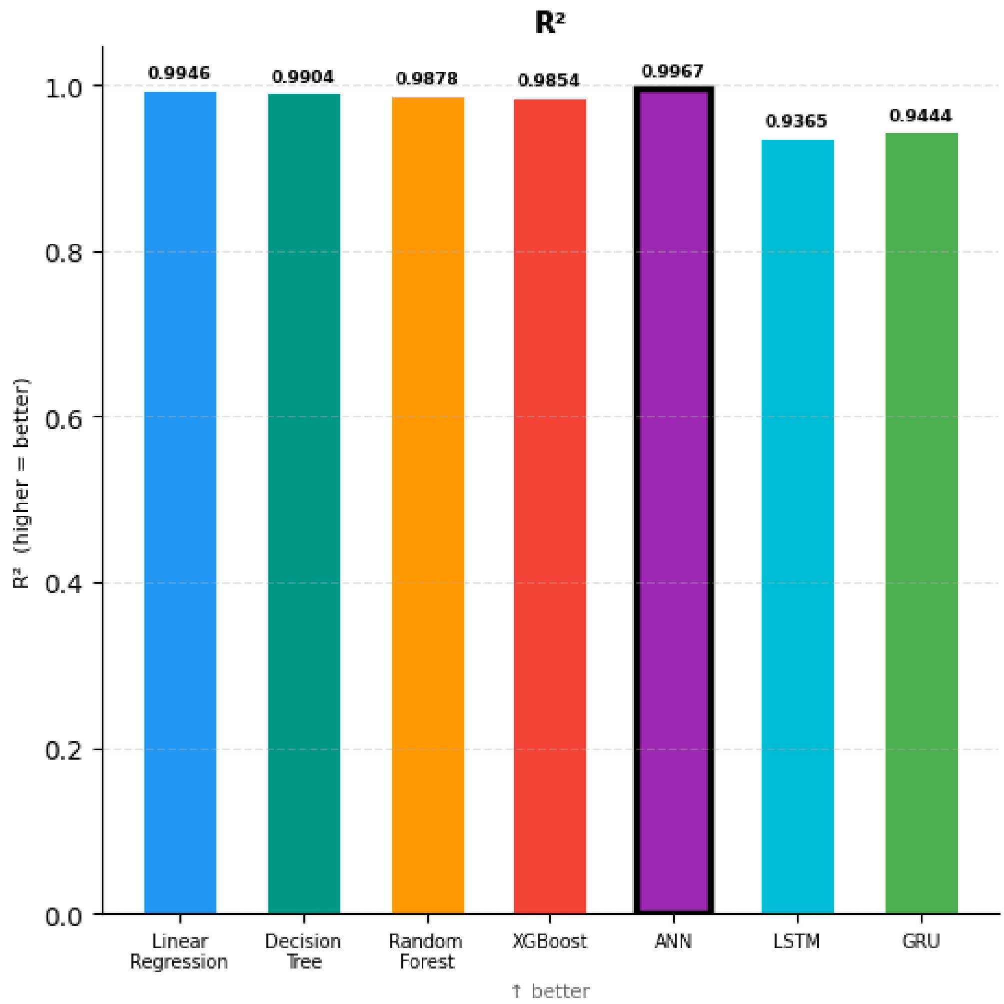


Figure 4.6: Coefficient of Determination (R^2) Score distribution for supervised machine learning and deep learning models.

Moreover, Figure 4.7 presents the evaluation metrics related to classification accuracy. ANN takes the lead yet again, with a score of 95.28% accuracy rate. It can deal with the precision of the dataset better than any other algorithm can.

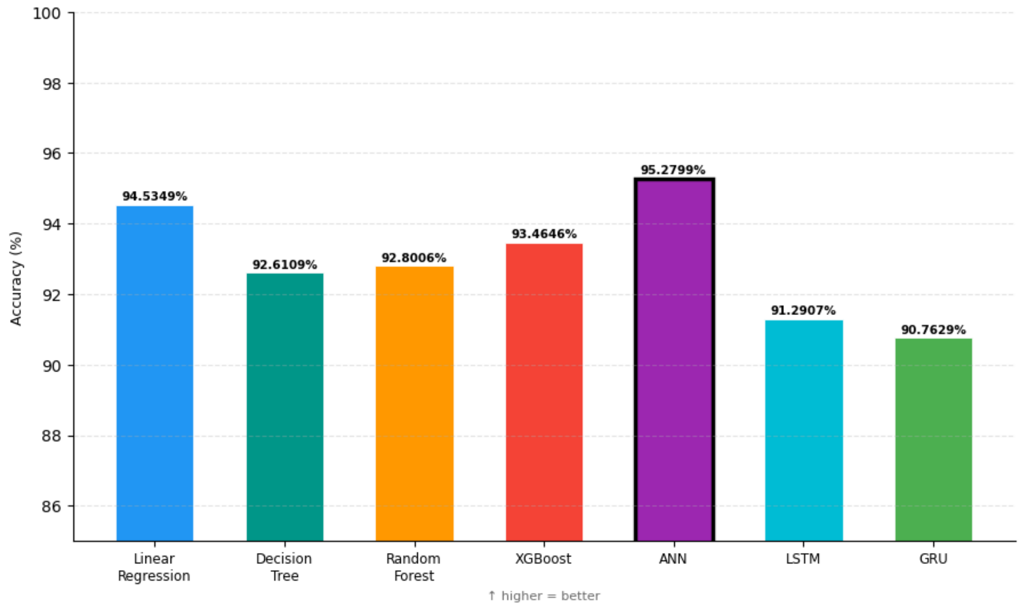


Figure 4.7: Model performance comparison based on prediction Accuracy (%).

On the other hand, Figure 4.8 illustrates the performance in terms of Weighted F1-Score. The ANN model achieved a score of 0.9234, demonstrating its robustness in capturing the complex charging patterns across the dataset compared to other models.

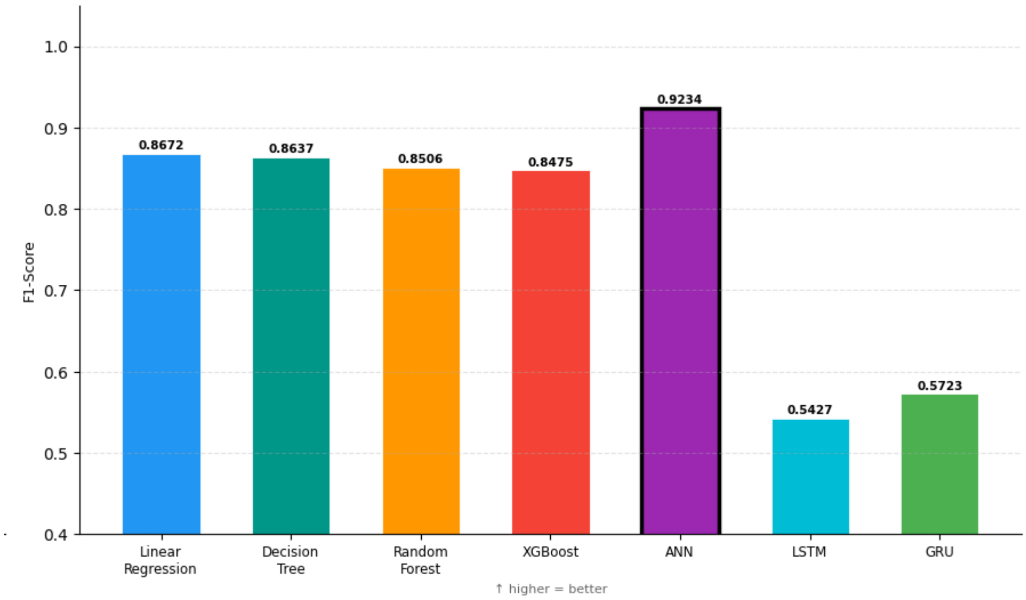


Figure 4.8: Weighted F1-Score distribution across different machine learning models.

4.3 Implications

The findings from this research give crucial insights into the topic areas of load prediction, smart charging facilities, and the urban energy management system using electric vehicles. They have implications for the management of the grid system, urban planning, and intelligent systems. Specifically, when applied to the urban setting of Khulna.

4.3.1 Grid and Charging Network Management

The highly accurate predictions of ML algorithms ($R^2 \approx 0.9967$) reveal that predicting the charging load at the level of the entire EV fleet based on simulated data is feasible. Thus, the power grid can apply a **proactive management strategy** that predicts the load before reaching its peak value. This strategy aligns well with the latest discoveries in integrated approaches to forecasting that consider the cooperation between simulations and to improve urban power grids [3]. Furthermore, the simulation demonstrates that smart charging policies combined with queuing reduce the burden of charging demands on charging stations. Clearly, integrating proactive techniques with simulation results yields better performance and reliability in the grid network.

4.3.2 Data-Driven Feature Insights

Through the examination of the results obtained using feature importance and correlation analyses, it becomes evident that EV load demands depend on only a few macro-level features—namely, the presence of charging events and their timings during a day. Instead of the micro-level characteristics specific to each individual car, the macro-level characteristics of the entire fleet appear to be more important for estimating EV load demand. As a result, it does not become necessary to record all the relevant features of each individual vehicle. The use of a limited number of features enables one to successfully measure EV load demands in a real-life setting. Features such as the state of charge of a battery, the speed, and movement can provide contextual information but do not play a significant role in determining EV load demands. Therefore, the analysis presented here offers an opportunity to make accurate forecasts using a small amount of information.

4.3.3 Urban Infrastructure Planning

It becomes evident that it is crucial to be thoughtful when choosing the locations for setting up the charging stations in cities since busy intersections and popular places attract lots of cars and act like a vacuum cleaner, sucking all the resources and potentially becoming severe obstacles. Besides, the increase in cars' number at such a point will not only make the process of movement much slower but will also affect the power network negatively. The developed framework enables one to conduct a series of virtual tests in order to minimize risks and find the most optimal location for the installation of charging stations (e.g., Khulna).

4.3.4 Simulation as a Digital Twin Framework

The sumo simulation environment is analogous to having a virtual replica of city traffic and charging infrastructure. Such a simulation could provide insight into the movement and charging processes involved in a dynamic way. This would enable the creation of quality data sets required for machine learning as well as the execution of different traffic and charging scenarios without any dangers involved and without ever hitting the road. This is especially helpful for regions that lack the actual data concerning electric vehicles.

4.3.5 Machine Learning for Load Forecasting

From a comparative analysis of the ML models, it becomes obvious that tree-based algorithms such as Decision Trees, Random Forests, and overshadow their counterparts—the linear models. is the algorithm that strikes the perfect balance between predictive power and generalization. It should be noted that the accuracy is extremely high, almost unreal, which means that the data have highly regular patterns. Summing up all the observations from this research, one can conclude that machine learning is an excellent technique to use in predicting structured EV loads. In line with this conclusion, further research shows that while tree-based models work well in structured environments, hybrid structures like CNN-LSTM-AM should be used to account for more complicated patterns [4].

4.3.6 Roadmap for Smart and Sustainable Electrification

In contrast to previous studies, this one proposes an operational model for electric vehicles deployment in developing cities. Simulations produce realistic data, from which machine learning-based forecasting is performed and intelligent charging systems with queuing design are developed. These processes combine to form the **Monitor-Predict-Control** framework that allows the development of control strategies for managing EV. Thus, institutions and policy makers receive opportunities for designing evidence-based policies for transport and energy sector development. This ensures that the integration of EV will not threaten the power grid operation.

Chapter 5

Conclusion

Chapter 5: Conclusion

This thesis proposes an integrated approach for predicting electric vehicle charging demands. Our approach involved the integration of a comprehensive traffic simulator together with supervised machine learning. We first started off with the use of SUMO and the TraCI API in Python, which is mainly used to control and simulate the entire road network. Our proposed model consists of ten configurable charging stations, numerous Electric Vehicles, and a queue management system with charging suggestions.

The simulated data had 34,020 entries and 25 attributes. Monitoring each vehicle separately and the entire system within the 600-second period. We applied seven different machine learning algorithms for this purpose, starting with Linear Regression up to highly complex Artificial Neural Network (ANN) model. The ANN model demonstrated the highest accuracy with R^2 = 0.9967. Surprisingly enough, the Linear Regression model managed to reach an equally impressive R^2 value of 0.9946. While the tree-based and recurrent network models gave reliable yet slightly less accurate predictions.

5.1 Concluding Remarks

What makes our analysis unique?

1. The construction of a highly efficient SUMO-based microscopic simulator [10] tailored specifically to handle the wide variety of vehicles used in Bangladesh – namely Easy Bikes, Electric Rickshaws, and Electric Vans.

2. The availability of an extensive dataset, which includes 34,020 data points and 25 dynamic variables. Sufficient enough to analyze and understand the use of energy and make proper load predictions.

3. Testing of seven different regression models, such as those using deep learning techniques, where the Artificial Neural Network (ANN) was found as the best performing short-term charging load forecasting model.

4. Development of the rule-based smart charging strategy, illustrating how real-time rerouting of vehicles according to SOC can be implemented to alleviate any potential grid congestion.

Simulation data can be seen from our findings to be very effective in cases where EV-based charging datasets are difficult to access, particularly in rapidly changing environments such as Bangladesh. In this study, we have provided policymakers and electricity companies with an approach based on data to follow during the transportation electrification era.

5.2 Limitations

In spite of having attained reliable findings on the prediction of EV load and simulation, there are various limitations attached to this research. These should be considered while assessing the findings and suggesting improvements for the future.

Reliance on Simulation Data: In our research, we have simulated the data using the SUMO environment rather than collecting any real-world data. We have done our best to keep this simulation as realistic as possible. The objective behind this is to incorporate all real-life scenarios along with the behavior of the electric vehicle in our model. But, a simulation can never be perfect and there are many unknowns that cannot be anticipated in the virtual world, such as how the driver will operate the car, unexpected traffic or unexpected charging behavior of the electric vehicle. Due to this, our model will definitely perform better in a test than in practice.

Feature Inadequacy: Though the data set includes features related to both system and vehicle level factors, there are certain real-life aspects that are not covered by the model. Some of these are : weather conditions, such as temperature or rain, which have no mention in the model. Behavior patterns of people while charging their vehicles, including their preferences and urgency, are left out. Electricity price dynamics are not considered and neither are infrastructural failure events accounted for.

Static Assumption on Infrastructure: In this case, the model is static, meaning the infrastructure remains unaltered during the simulation, and the charging facilities are fixed. However, this does not happen in reality. The infrastructure is flexible and subject to change, where the charging facilities could be closed down, new charging facilities created, among other factors which might influence the distribution of the load.

5.3 Recommendations for Further Research

To facilitate further expansion of the research scope and relevance, several suggestions for future works are given below:

Inclusion of Real-Life Data: In order to ensure validity and applicability in practice, researchers have to rely heavily on real-life EV charging and traffic data. The SUMO simulation provides control and precision, yet, lacking a significant amount of real-life data, fails to cover many critical points. The actual traffic situation tends to be unpredictable due to multiple reasons: driver's behaviour is highly irregular, current infrastructure is often not able to meet the current needs, demand may vary significantly in a variety of ways that might go unnoticed in simulations. Including data from EV stations, smart metering systems, or transportation agencies will allow the model of the future to benefit from practical experience, making the prediction much more complicated, but also realistic and applicable. It will greatly enhance applicability of the developed model by connecting it to the real life and allowing integration into smart grid applications (in countries such as Bangladesh).

Deep Learning Architectures: While this paper explored the application of deep learning architectures such as ANN, LSTM, and GRU, there is some room left for advancement. The next phase of research may explore even more sophisticated hybrid networks such as CNN-LSTM-AM [4]. This type of model is capable of detecting spatial and multi-level temporal dependencies. This is particularly important when it comes to EV charging because the demand varies during the day. In order to understand the dynamics of the load demand better, one should feed the model with the historical information and let the deep learning do the rest of the work.

Intelligent Charging Optimization: In this paper, a simple intelligent charging system based on rules is implemented. This provides a simple technique for allocating the energy demand of different vehicles among several charging stations. However, there exists a lot of room for improvement, where more advanced optimization techniques can be applied. Future works may include more sophisticated approaches like reinforcement learning, genetic algorithms or multi-objective optimization, which provide a self-learning mechanism. Such intelligent systems can automatically decide where to allocate each vehicle in terms of charging, based on real-time data related to traffic conditions, station occupancy and power grid limitations. This will lead to better load management and lower peak demands on the power grid.

Expansion of the Simulation Environment: In order to make this study more generalized and realistic, the future research must focus on making some enhancements in the simulation environment like including- large scale road networks, junctions, and traffic controls systems. Also, incorporating different kinds of vehicles having different behavioral patterns is something that must be included in an improved model of this study. Incorporating random elements like-sudden traffic congestions, accidents, or unexpected increases in the demand will make the simulation more realistic. Through these improvements, it will become easier for the researchers to analyze the performance of the load forecasting models under various circumstances.

Renewable Energy Integration: The incorporation of renewable energy sources into the system of EV charging stations could also be considered for future study on this topic. This would include the creation of a model where EV charging stations use solar panels or hybrid energy systems for their operation. Another potential direction could involve the study of the efficient utilization of renewable energy generation to satisfy the charging needs. In this case, studying the time frame for balancing generation and consumption as well as energy storage techniques would be required. Through adding renewable energy to the model, transportation would become sustainable.

Deployment of Real-Time Systems: The final point for future work is that there should be efforts made towards deployment of the systems in real-time. This involves incorporating machine learning algorithms within the Internet of Things (IoT) enabled Smart Grids where data is constantly collected and analyzed. Using efficient frameworks such as DeepBoost [3] may prove to be beneficial in terms of real-time monitoring applications. Deployment of these real-time systems will involve tracking the demands for charging, forecasting trends and managing resources through automation. With these measures, decision making for service providers and policy makers will be greatly assisted. Besides, it will enable bridging the gap between theoretical research and practical transport and energy management.

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